Chapter 4:

Advanced Derivative Applications

Chapter 4 Overview:

There are two main contexts for derivatives: graphing and motion. In this chapter, we will consider both these applications of the derivative. Much of this is a review of material covered in precalculus. Key topics include:

- Extremes
- The First and Second Derivative Tests
- Optimization
- Derivative-related theorems
- Intervals of increasing and decreasing
- Advanced graphing using the derivative
- Rectilinear motion

This chapter makes up a solid portion of the multiple choice, along with also being at least one full free response question (and often parts of others) on the AP exam.

4.1: Extrema and the Derivative Tests

One of the most valuable aspects of calculus is that it allows us to find extreme values of functions. The ability to find maximum or minimum values of functions has wide-ranging applications. Every industry has uses for finding extremes, from optimizing profit and loss, to maximizing output of a chemical reaction, to minimizing surface areas of packages. This one tool of calculus eventually revolutionized the way the entire world approached every aspect of industry. It allowed people to solve formerly unsolvable problems.

OBJECTIVES

Find Critical Values and Extreme Values for Functions.
Use the 1^{st} and 2^{nd} Derivative Tests to Identify Maxima v.s. Minima.

But, before we begin, let's review some definitions from last year, which will be useful in this chapter as well as subsequent chapters.

Critical Value → The x-coordinate of the extreme.

Maximum Value → The y-coordinate of the high point.

Minimum Value → The y-coordinate of the low point.

Relative Extremes → The highest or lowest point in a given section of the curve.

Absolute Extremes → The highest or lowest point of the whole curve.

Interval of Increasing → The interval of x-values for which the curve is rising from left to right.

Interval of Decreasing → The interval of x-values for which the curve is dropping from left to right.

Critical values of a function occur when:

1) $\dfrac{dy}{dx} = 0$

2) $\dfrac{dy}{dx}$ does not exist

3) the function reaches an endpoint of its domain

It's also helpful to remember that a **critical value** is referring specifically to the value of the x, while the **extreme value** refers to the value of the y.

As we already know, the first derivative is what allows us to algebraically find extremes, and the First Derivative Test allows us to interpret critical values as maxima or minima. Since the first derivative of a function tells us when that function is increasing or decreasing, we can figure out if a critical value is associated with a maximum or minimum depending on the sign change of the derivative. Let's quickly refresh our memory on the First Derivative Test.

The First Derivative Test

As the sign pattern of the first derivative is viewed left to right, the critical value represents a:

1) relative maximum if the sign changes from $+$ to $-$

2) relative minimum if the sign changes from $-$ to $+$

3) neither a relative max or min if the sign does not change

Ex 4.1.1: Apply the First Derivative Test to the function $y = 5x^4 - 10x^2$.

Sol 4.1.1:

$$\frac{dy}{dx} = 20x^3 - 20x = 0$$

$$20x\left(x^2 - 1\right) = 0$$

$$x = -1,\ 0,\ 1$$

Now, let's make our sign pattern.

y'	$-$	0	$+$	0	$-$	0	$+$
x		-1		0		1	

So, there appear to be critical values at x values of -1, 0, and 1. By utilizing the First Derivative Test, we can see that we have a $\boxed{\text{minimum at } -1}$, a $\boxed{\text{maximum at } 0}$, and another $\boxed{\text{minimum at } 1}$.

Ex 4.1.2: Find the maximum values for the function $g(x) = \sqrt{x^3 - 9x}$.

Sol 4.1.2: For this question, we have to be careful! Because we are dealing with a square root, we will have domain restrictions. We can find such restrictions by starting with a sign pattern for $g(x)$.

$$\begin{array}{c|ccccccc} g(x) & - & 0 & + & 0 & - & 0 & + \\ \hline x & & -3 & & 0 & & 3 & \end{array}$$

Now, since we know that the argument of a square root has to be positive, we know that our domain is when $x^3 - 9x \geq 0$. Therefore, based on our sign pattern, our domain is $x \in [-3,\, 0] \cup [3, \infty)$.

With that out the way, we can now take the first derivative and apply the First Derivate Test.

$$g'(x) = \frac{3x^2 - 9}{2\sqrt{x^3 - 9x}}$$

$$g'(x) = 0 \text{ when } 3x^2 - 9 = 0 \therefore x = \pm\sqrt{3}$$

$$g'(x) = DNE \text{ when } x^3 - 9x = 0 \therefore x = \pm 3,\, 0$$

Since $x = \sqrt{3}$ is not in the domain of the function, our critical values are at $x = \pm 3,\, 0, -\sqrt{3}$. Therefore, our sign pattern looks like:

$$\begin{array}{c|ccccccccc} g'(x) & DNE & + & 0 & - & DNE & & & DNE & + \\ \hline x & -3 & & -\sqrt{3} & & 0 & & \sqrt{3} & 3 & \end{array}$$

From our sign pattern, we can conclude that at $x = -\sqrt{3}$, we have a maximum value, and by substituting this value into the function, we find the maximum value is at $\boxed{y = 3.224}$.

The First Derivative Test is not the only way to test whether critical values are associated with maximum or minimum values. If you recall, the second derivative can show us intervals of concavity. This is very useful because if we know that the curve is concave down at a critical value, it must be associated with a maximum. Similarly, if the curve is concave up, the critical value must be associated with a minimum value.

The Second Derivative Test

For a function f:

1) If $f'(c) = 0$ and $f''(c) > 0$, then f has a relative minimum at c

2) If $f'(c) = 0$ and $f''(c) < 0$, then f has a relative maximum at c

Now, we can revisit **Ex 1.5.7** part (c), which we previously could not do without the Second Derivative Test.

Ex 4.1.3: Consider the curve given by $x^2 + 4xy + y^2 = -12$.

(c) Find the value(s) of $\dfrac{d^2y}{dx^2}$ at the point(s) found in part (b). Does the curve have a local maximum, a local minimum, or neither at those points? Justify your answer.

(Ex 1.5.7)

Sol 4.1.3:
(c) What is really asked here is to apply the Second Derivative Test, because we cannot create a sign pattern for non-functions. The y is not isolated in the equation. Recall that the two points we found in part (b) were $(-4,\ 2)$ and $(4,\ -2)$.

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left[\frac{dy}{dx} = -\frac{x+2y}{2x+y}\right]$$

$$= \frac{(2x+y)\left(1+2\dfrac{dy}{dx}\right) - (x+2y)\left(2+\dfrac{dy}{dx}\right)}{(2x+y)^2}$$

$$\left.\frac{d^2y}{dx^2}\right|_{(-4,2)} = \frac{(2(-4)+2)\,(1+2(0)) - (-4+2(2))\,(2+0)}{(2(-4)+2)^2}$$

$$= \frac{6}{(-6)^2} > 0$$

$$\left.\frac{d^2y}{dx^2}\right|_{(4,-2)} = \frac{(2(4)-2)\,(1+2(0)) - (4+2(-2))\,(2+0)}{(2(4)+2)^2}$$

$$= -\frac{6}{(-6)^2} < 0$$

$\boxed{(-4, 2) \text{ will be a minimum}}$ because the second derivative is positive.

$\boxed{(4, -2) \text{ will be a maximum}}$ because the second derivative is negative.

Ex 4.1.4: Use the Second Derivative Test to determine if the critical values of $g(t) = 27t - t^3$ are at a maximum or minimum value.

Sol 4.1.4:

$$g'(t) = 27 - 3t^2 = 0$$

$$t = \pm 3$$

So, -3 and 3 are critical values. Now, let's take a look at the second derivative of $g(t)$ at those values.

$$g''(t) = -6t$$

$$g''(-3) = -6(-3) = 18$$

$$g''(3) = -6(3) = -18$$

Finally, we can apply the Second Derivative Test. Since $g'(-3) = 0$ and $g''(-3) > 0$, we can determine that $\boxed{t = -3 \text{ is a minimum}}$. Similarly, since $g'(3) = 0$ and $g''(3) < 0$, we can determine that $\boxed{t = 3 \text{ is a maximum}}$. Note that the numerical value of the second derivative is irrelevant — only the sign matters.

Ex 4.1.5: Determine if the critical values of $y = \left(3 - x^2\right) e^x$ are at a maximum or a minimum value.

Sol 4.1.5: First, let's find the critical values.

$$\frac{dy}{dx} = \left(3 - x^2\right) e^x + e^x(-2x) = 0$$

$$-e^x \left(x^2 + 2x - 3\right) = 0$$

$$-e^x(x+3)(x-1) = 0$$

From here, we can see that our critical values are $x = -3, 1$. The derivative is never DNE. So, let's find the second derivative and substitute in our values.

$$\frac{dy}{dx} = -e^x \left(x^2 + 2x - 3\right)$$

$$\frac{d^2y}{dx^2} = -e^x(2x+2) + \left(x^2 + 2x - 3\right)(-e^x)$$

$$= -e^x \left(x^2 + 4x - 1\right)$$

$$\left.\frac{d^2y}{dx^2}\right|_{x=1} = -4e$$

$$\left.\frac{d^2y}{dx^2}\right|_{x=-3} = 4e^{-3}$$

By the Second Derivative Test, we can determine that $\boxed{x = 1 \text{ is a maximum}}$ and $\boxed{x = -3 \text{ is a minimum}}$.

Optimization

Optimization is a practical application of finding maxima and minima for functions. As stated before, this idea revolutionized thinking and is a critical component of all industries. We've previously approached this topic in precalculus, but now, this concept forms a much more fundamental part of what we need to be able to do.

OBJECTIVES

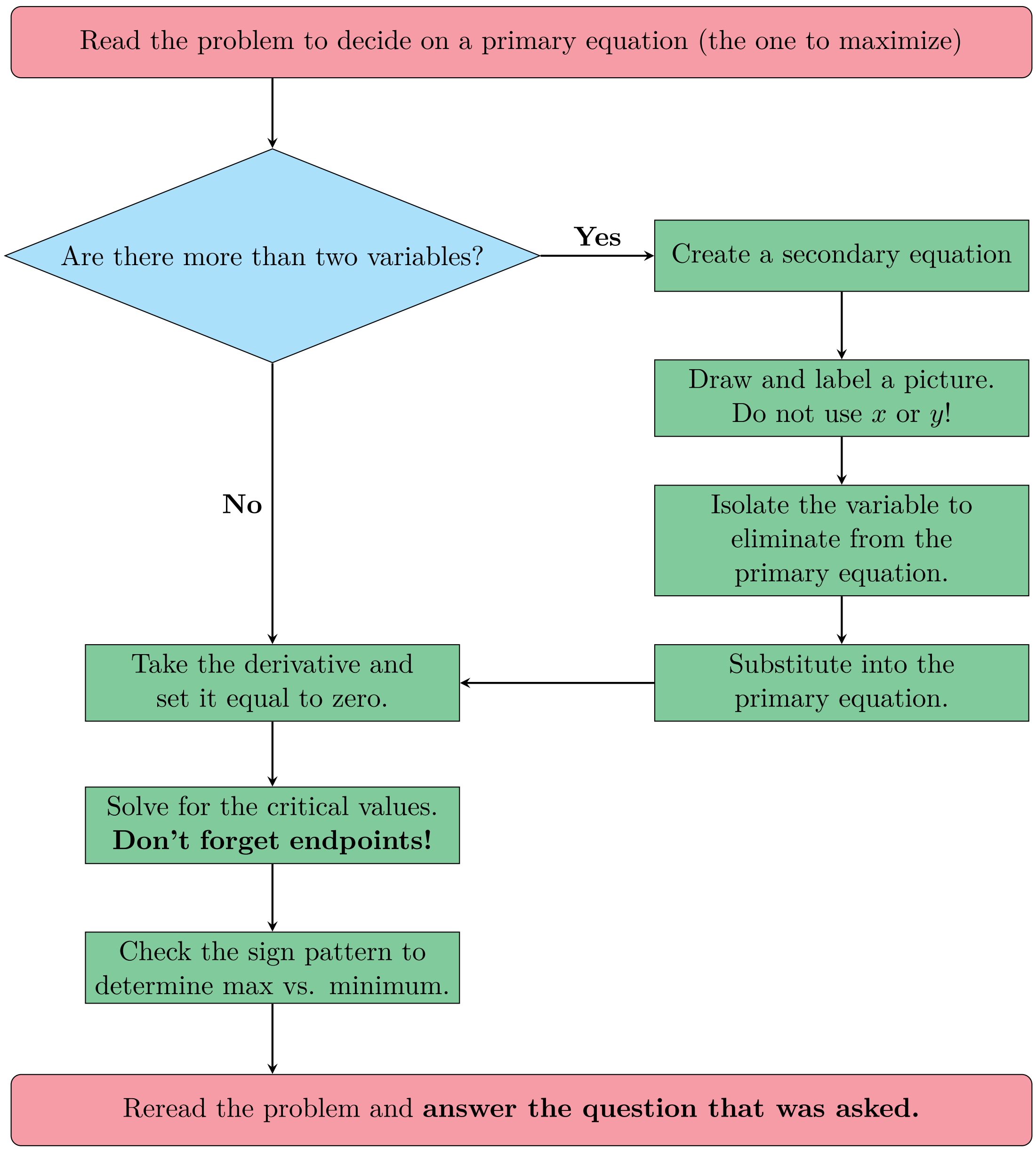
Solve Optimization Problems.

Every optimization problem looks a bit different, but they all follow a similar progression. You must first identify your variables and any formula you need. Use algebra to eliminate any variables, and take the derivative of **the function you're trying to optimize**. That is the most common mistake in optimization: taking the derivative of the wrong function. Now, let's take a look at the following example:

Ex 4.1.6: The volume of a cylindrical cola can is 32π in^3. What is the minimum surface area for such a can?

The word "minimum" tells us that we have an optimization problem. Recall our workflow for tackling optimization problems:

Strategy For Optimization Problems



So, let's tackle our example.

Sol 4.1.6: The problem asks to minimize surface area, which is determined by:

$$S = 2\pi r^2 + 2\pi r h$$

As there are more than two variables in this equation, either r or h needs to be eliminated in this formula before differentiating. The volume is $V = \pi r^2 h = 32\pi$, so $h = \dfrac{32}{r^2}$ and

$$S = 2\pi r^2 + 2\pi r\left(\frac{32}{r^2}\right)$$

$$= 2\pi r^2 + \frac{64\pi}{r}$$

$$S' = 4\pi r - \frac{64\pi}{r^2} = 0$$

$$\therefore 4\pi r = \frac{64\pi}{r^2}$$

$$r^3 = 16 \rightarrow r = 2.5198$$

Now, let's make a sign pattern to determine if this critical value is a minimum.

$$\begin{array}{ccccc} S' & - & 0 & + \\ & \longleftarrow & \longrightarrow & \\ r & & 2.520 & \end{array}$$

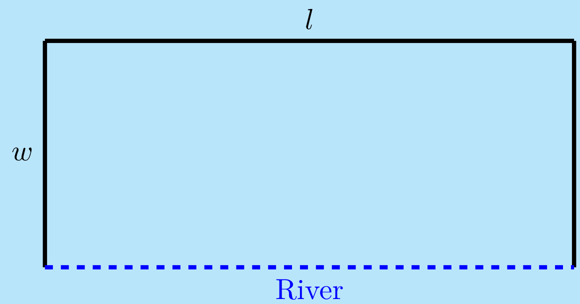
Because the derivative changes from negative to positive, we know that 2.5198 is a minimum. We can plug it back into our surface area equation to find our minimum surface area.

$$S(2.5198) = 2\pi(2.5198)^2 + \frac{64\pi}{2.5198}$$

$$S(2.5198) = 119.687 \text{ in}^2$$

Therefore, the minimum surface area of the cola can is $\boxed{119.687 \text{ in}^2}$.

Ex 4.1.7: The owner of the Rancho Grande has 3000 yards of fencing material with which to enclose a rectangular piece of grazing land along the straight portion of a river. If fencing is not required along the river, what are the dimensions of the largest area he can enclose? What is the area?



Sol 4.1.7: We first begin with our area formula, $A = lw$. Now, as we know, we need to express one variable in terms of the other. We can do this by utilizing the fact about perimeter to complete the problem.

$$P = l + 2w$$

$$3000 = l + 2w$$

$$3000 - 2w = l$$

Let's substitute this back into our area formula.

$$A = (3000 - 2w)w$$

$$A = 3000w - 2w^2$$

Now, since we have an equation with one independent variable, we can take the derivative easily.

$$\frac{dA}{dw} = 3000 - 4w = 0$$

$$w = 750$$

$$l = 1500$$

So, we would want a $\boxed{\text{width of 750 yards}}$ and a $\boxed{\text{length of 1500 yards}}$. This would give us an area of $\boxed{1,125,000 \text{ yd}^2}$.

Ex 4.1.8: Find the point on the curve $y = \dfrac{e^{-x^2}}{2}$ that is closest to the origin.

Sol 4.1.8: We want to minimize the distance to the origin, so we will be using the Pythagorean Theorem to find the distance.

$$D = \sqrt{x^2 + y^2}$$

$$D = \sqrt{x^2 + \left(\frac{e^{-x^2}}{2}\right)^2}$$

$$\frac{dD}{dx} = \frac{1}{2}\left(x^2 + \frac{e^{-2x^2}}{4}\right)^{-\frac{1}{2}} \left(2x - xe^{-2x^2}\right) = 0$$

$$x = 0,\ \pm 0.841$$

$$\begin{array}{c|ccccccc} \frac{dD}{dx} & - & 0 & + & 0 & - & 0 & + \\ \hline x & & -0.841 & & 0 & & 0.841 & \end{array}$$

By the First Derivative Test, we know that $x = 0$ is a minimum of the curve. Plugging that value back into our original equation gives us our desired point of $\boxed{(0,\ 0.5)}$.

4.1 Free Response Homework

Find the critical and extreme values for each function.

1. $y = 2x^3 + 9x^2 - 168x$

2. $y = 3x^4 + 2x^3 + 12x^2 + 12x - 42$

3. $y = \dfrac{x^2 + 1}{x^3 - 4x}$

4. $y = \dfrac{x - 5}{x^2 + 9}$

5. $y = \sqrt{9x^3 - 4x^2 - 27x + 12}$

6. $y = \sqrt{\dfrac{3x}{9 - x^2}}$

7. $y = x^2 \sqrt[3]{9 - x^2}$

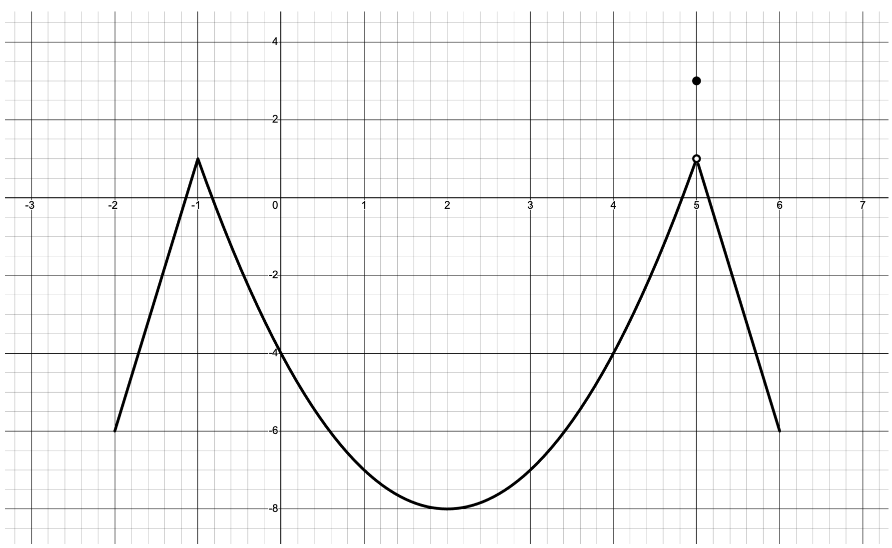
8. $y = \left(x - x^2\right) 3^x$

9. $y = \dfrac{\sqrt{3}}{2}x + \cos(x) \quad \Big| \quad x \in [-2\pi,\, 2\pi]$

10. $y = \cot^{-1}\left(\dfrac{1}{x}\right) - \tan^{-1}(x)$

11. $y = \ln\left(x^2 + 4x + 14\right)$

12.



For 13 – 18, find the absolute maximum and minimum values of f on the given intervals.

13. $f(t) = \dfrac{x}{x^2 + 1} \quad \Big| \quad x \in [0,\, 2]$

14. $f(t) = \sqrt[3]{t}(8 - t) \quad \Big| \quad t \in [0,\, 8]$

15. $f(z) = ze^{-z} \quad \Big| \quad z \in [0,\, 2]$

16. $f(x) = \dfrac{\ln x}{x} \quad \Big| \quad x \in [1,\, 3]$

17. $f(x) = e^{-x} - e^{-2x}$ $\quad\Big|\quad$ $x \in [0,\, 1]$

18. $f(y) = \sin\left(\dfrac{y}{3}\right) + \dfrac{2y}{9}$ $\quad\Big|\quad$ $y \in [-10,\, 15]$

For each of the following functions, apply the First Derivative Test, then verify the results with the Second Derivative Test.

19. $f(x) = x^2 \ln x$

20. $h(f) = f^3 - 12f + 21$

21. $f(x) = xe^{-x^2}$

22. $h(t) = e^{t^3+t^2+1}$

Solve these problems algebraically.

23. Find two positive numbers whose product is 110 and whose sum is a minimum.

24. Find a positive number such that the sum of the number and its reciprocal is a minimum.

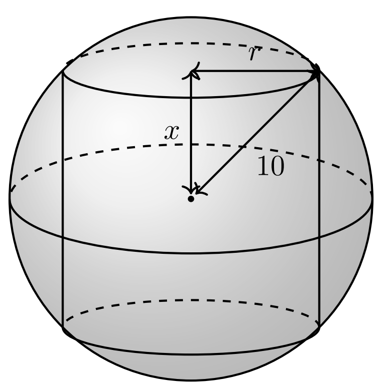
25. A farmer with 750 feet of fencing material wants to enclose a rectangular area and divide it into four smaller rectangular pens with sides parallel to one side of the rectangle. What is the largest possible total area.

26. If 1200 cm^2 of material is available to make a box with an open top and a square base, find a maximum volume the box can contain.

28. Find the point(s) on the curve $y = \dfrac{1}{x^2+1}$ that is/are closest to the origin.

29. Find the area of the largest rectangle that can be inscribed in the ellipse $\dfrac{x^2}{16} + \dfrac{y^2}{9} = 1$.

30. Find the maximum volume of a cylinder inscribed in a sphere with radius 10.



31. A piece of wire 10 m long is cut into two pieces. One piece is bent into a square, while the other is bent into an equilateral triangle.

(a) Find where the wire should be cut to maximize the area enclosed.

(b) Find where the wire should be cut to minimize the area enclosed.

32. You need to enclose 500 cm^3 of fluid in a cylinder. If the material you are using costs 0.001 dollars per cm^3, find the size of the cylinder that minimizes the cost. If the product that you are containing is a sports drink, do you think that the size that minimizes the cost is the most efficient size? Explain

33. The height of a man jumping off of a high dive is given by the function $h(t) = -4.9t^2 + 2t + 10$ on the domain $0 \leq t \leq 1.64715$. Find the absolute maximum and minimum heights modeled by this function.

34. Given the area of a triangle can be calculated with the formula $A = \frac{1}{2}ab\sin(\theta)$, what value of θ will maximize the area of a triangle (given that a and b are constants)?

35. You operate a tour service that offers the following rates for the tours: \$200 per person if the minimum number of people book the tour (50 people is the minimum) and for each person past 50, up to a maximum of 80 people, the cost per person is decreased by \$2. It costs you \$6000 to operate the tour plus \$32 per person.

(a) Write a function that represents cost, $C(x)$.

(b) Write a function that represents revenue, $R(x)$.

(c) Given that profit can be represented as $P(x) = R(x) - C(x)$, write a function that represents profit and state the domain for the function.

(d) Find the number of people that maximizes your profit. What is the maximum profit?

4.1 Multiple Choice Homework

1. Give the value of x where the function $f(x) = x^3 - 9x^2 + 24x + 4$ has a relative maximum point.

a) 4 b) -2 c) 2 d) -4 e) 3

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2. Give the value of x where the function $f(x) = x^3 - \frac{33}{2}x^2 + 84x - 2$ has a relative minimum point.

a) -4 b) -7 c) 4 d) 5 e) 7

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3. Give the approximate location of a relative maximum point for the function $f(x) = 3x^3 + 5x^2 - 3x$.

a) $(-1.357, 5.779)$ b) $(0.2457, -0.3908)$ c) $(-1.357, 5.713)$ d) $(0.2457, -0.3216)$ e) $(-1.357, -0.3908)$

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4. Consider the following sign pattern:

$\frac{dy}{dx}$	$+$	0		0	$-$	0	$+$	0		0	$-$
x		-2		$-\sqrt{2}$		0		$\sqrt{2}$		2	

What value of x does y has a local minimum?

a) -2 b) $-\sqrt{2}$ c) 0 d) $\sqrt{2}$ e) 2

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5. Find the absolute maximum of $y = -\sqrt{25 - x^2}$ on the interval $x \in [-2, 4]$.

a) -2 b) 0 c) -5 d) $-\sqrt{21}$ e) -3

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6. Find the absolute maximum value of $y = \sqrt{36 - x^2}$ on the interval $x \in [-2, 2]$.

a) 5 b) 6 c) 7 d) 0 e) 1

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7. The graph of the function $f(x) = 2x^{\frac{5}{3}} - 5x^{\frac{2}{3}}$ is increasing on which of the following intervals.

I. $1 < x$ II. $0 < x < 1$ III. $x < 0$

a) I only b) II only c) III only d) I and II only e) II and III only

8. If $f(x) = x^2 e^{-2x}$, then the graph of f is increasing for all x such that

a) $0 < x < 1$ b) $0 < x < \frac{1}{2}$ c) $0 < x < 2$ d) $x < 0$ e) $x > 0$

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9. If $f(x) = \dfrac{x}{\ln(7x)}$, then what is the interval of decreasing?

a) $\left(0, \frac{1}{7}\right)$ b) $\left(0, \frac{1}{7}\right) \cup \left(\frac{1}{7}, \frac{1}{7}e\right)$ c) $(1, 7e)$ d) $(1, 7)$ e) $\left(1, \frac{1}{7}e\right)$

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10. Suppose $f'(x) = \dfrac{(x+1)^3(x-4)^2}{(x^4+1)}$. Which of the following statements must be true?

I. The slope of the line tangent to $y = f(x)$ at $x = 1$ is 36.

II. $f(x)$ is increasing on $x \in (1, 4)$.

III. $f(x)$ has a minimum at $x = -1$.

a) II only b) I and III only c) II and III only d) I and II only e) I, II, and III

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11. Consider the following sign pattern:

$f'(x)$	$-$	0	$+$	0	$-$	0	$-$
x		-4		-1		2	

a) -4 b) -1 c) 1 d) 2 e) None of these

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12. The sum of two positive integers is x and y is 150. Find the value of x that minimizes $P = x^3 - 150xy$.

a) $x = 25$ b) $x = 50$ c) $x = 75$ d) $x = 100$ e) $x = 125$

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13. The sum of two positive integers x and y is 120. Find the value of x that minimizes $P = x^3 - 120xy$.

a) $x = 20$ b) $x = 40$ c) $x = 60$ d) $x = 80$ e) $x = 100$

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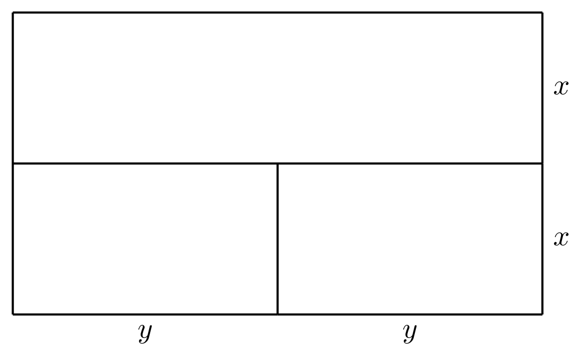
14. A farmer has 100 yards of fence to enclose a field, subdivide it into two equal pens, and further subdivide one of those pens into two equal fields as shown below.



a) 12.5 b) 10 c) $\dfrac{100}{11}$ d) $\dfrac{25}{3}$ e) None of these

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15. A farmer with 890 ft of fencing wants to enclose a rectangular area and then divide it into four pens with fencing parallel to one side of the rectangle. What is the largest possible total area of the four pens?

a) $19,825.5$ ft^2 b) $19,805.5$ ft^2 c) $19,801.5$ ft^2 d) $19,902.5$ ft^2 e) $19,791.5$ ft^2

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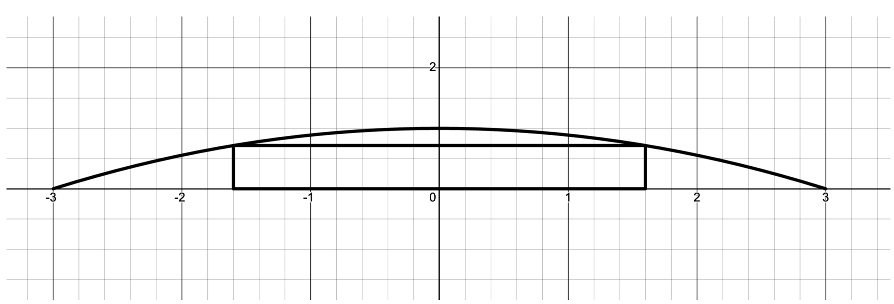
16. A rectangle with one side on the x-axis has its upper vertices on the graph of $y = 1 - \dfrac{x^2}{9}$, as shown in the figure below. What is the maximum area of the rectangle?



a) $\sqrt{3}$ b) $\frac{\sqrt{3}}{2}$ c) $\frac{2}{3}$ d) $\frac{4\sqrt{3}}{3}$ e) None of these

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17. In a certain community, an epidemic spreads in a way that the percentage P of the population that is infected after t months is modeled by

$$P(t) = \frac{kt^2}{(C + t^2)^2},$$

where C and k are constants. Find t, such that P is maximized.

a) 0 b) $\sqrt{C}$ c) $\sqrt{k}$ d) $\sqrt{\frac{C}{3}}$ e) None of these

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18. Consider the following table below:

x	$g(x)$	$g'(x)$	$g''(x)$	$f(x)$	$f'(x)$	$f''(x)$
0	2	3	-2	2	1	-2
2	4	2	0	4	0	3
4	1	-1	3	0	2	-1

At $x = 2$, $f(x)$ has a:

a) Local min. b) Local max. c) Point of inflection d) Zero e) None of these

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19. Let $f(x)$ and $g(x)$ be twice-differentiable functions with selected values given in the table in **question 18**. Let $h(x) = g(f(x))$. At $x = 2$, $h(x)$ has a:

a) Local min. b) Local max. c) Point of inflection d) Zero e) None of these

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4.2: The Mean Value and Rolle's Theorem

The Mean Value Theorem is an interesting piece of the history of calculus that was used to prove a lot of what we take for granted. The Mean Value THeorem was used to prove that a derivative being positive or negative told you that the function was increasing or decreasing, respectively. Of course, this led directly to the First Derivative Test and the intervals of concavity.

Mean Value Theorem

If f is a function that satisfies these two hypotheses:

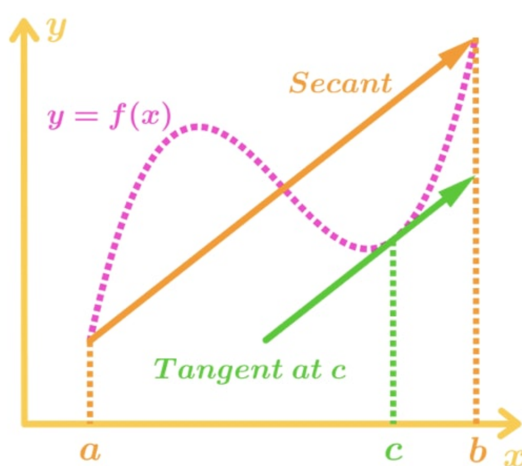
1) f is continuous on the closed interval $[a,\, b]$.

2) f is differentiable on the closed interval $(a,\, b)$.

Then there exists a number c in the interval $(a,\, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Again, translating from math to english, this simply states that if you have a smooth, continuous curve, the slope of the line connecting the endpoints has to equal the slope of a tangent somewhere in that interval. Alternatively, it says that the secant line (a line through two distinct points on a curve) through the endpoints will have the same slope as a tangent line at some point on the curve.



Ex 4.2.1: Show that the function $f(x) = x^3 - 4x^2 + 1$ with $x \in [-1,\, 3]$ satisfies all the conditions of the Mean Value Theorem and find c.

Sol 4.2.1: Polynomials are continuous throughout their domain, so the first hypothesis is satisfied. Polynomials are also differentiable throughout their domain, so the second condition is satisfied.

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$f'(c) = 3c^2 - 8c = \frac{f(3) - f(-1)}{3 - (-1)}$$

$$3c^2 - 8c = \frac{-8 - (-4)}{4}$$

$$3c^2 - 8c = -1 \implies 3c^2 - 8c + 1 = 0$$

$$c = \boxed{2.535,\ 0.131}$$

For this problem, we actually find *two* values of c. This is definitely possible, as the Mean Value Theorem doesn't state how many numbers c that satisfy the theorem; simply that there exists at least one number c.

Ex 4.2.2: Show that the function $f(x) = x^2 - 4x + 1$ with $x \in [0,\ 4]$ satisfies all the conditions of the Mean Value Theorem and find c.

Sol 4.2.2: Polynomials are continuous throughout their domain, so the first hypothesis is satisfied. Polynomials are also differentiable throughout their domain, so the second condition is satisfied.

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$f'(c) = 2c - 4 = \frac{f(4) - f(0)}{4 - 0}$$

$$2c - 4 = \frac{1 - 1}{4}$$

$$2c - 4 = 0$$

$$c = \boxed{2}$$

Now, let's talk a little about Rolle's Theorem. Rolle's Theorem is a specific case of the Mean Value Theorem (though Joseph-Louis Lagrange used it to prove the Mean Value Theorem).

Rolle's Theorem is extremely powerful, as it was used to prove all of the rules we have used to interpret derivatives. It used to be a very useful theorem, but is now something of a historical curiosity.

Rolle's Theorem

If f is a function that satisfies these three hypotheses:

1) f is continuous on the closed interval $[a,\, b]$.

2) f is differentiable on the closed interval $(a,\, b)$.

3) $f(a) = f(b)$

Then there exists a number c in the interval $(a,\, b)$ such that $f'(c) = 0$.

Although this theorem may seem confusing at first, it is simply stating that if you have a continuous and smooth curve with an initial point and ending point at the same height, then there exists some point in the curve that has a derivative of zero. This can be simply proved by utilizing the Mean Value Theorem and noticing that $f(b) - f(a) = 0$.

$$f'(c) = \frac{f(b) - f(a)}{b - a} \to f'(c) = \frac{0}{b - a} \implies f'(c) = 0$$

Another theorem that is important to note and often confused with Rolle's Theorem or the Mean Value Theorem is the **Extreme Value Theorem**. Recall what the Extreme Value Theorem states:

Extreme Value Theorem

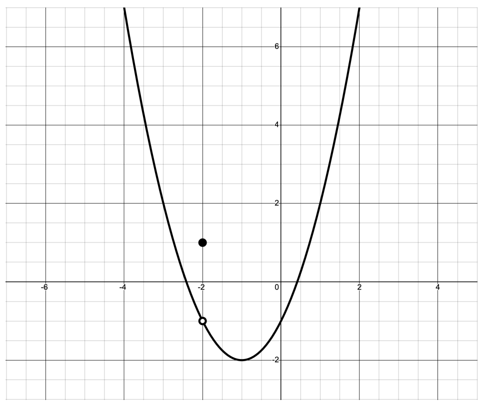
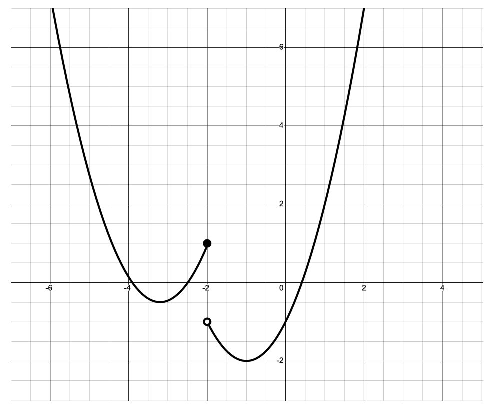
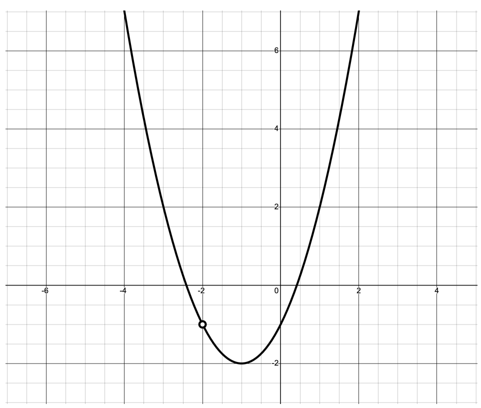
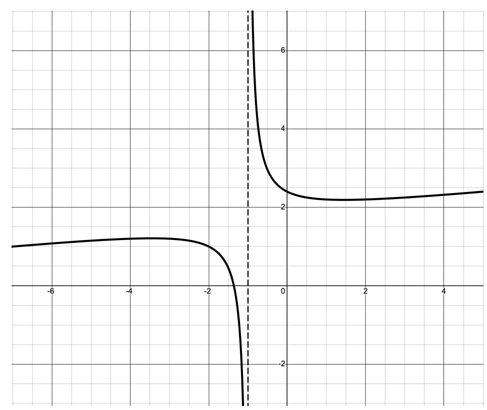
If f is continuous on the closed interval $[a,\, b]$, then $f(x)$ attains a maximum and a minimum at least once each.

The final theorem that may be confused with the Mean Value Theorem is the **Intermediate Value Theorem**. Recall what the Intermediate Value Theorem states:

Intermediate Value Theorem

If f is continuous on the closed interval $[a,\, b]$, then $f(x)$ attains every height (or value) between $f(a)$ and $f(b)$.

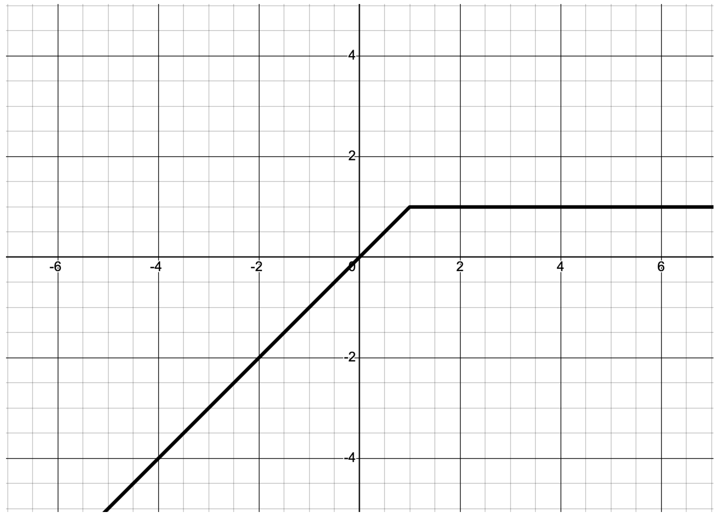
Now that we've gone over the Mean Value Theorem, we need to consider when this theorem might not apply to a problem — that is, when is a function not continuous or not differentiable. We first saw the concept of non-continuous functions in precalculus, but let's formalize these discontinuities within the following table.

	Removable Discontinuity $\left(\lim_{x \to a} f(x) \text{ Exists} \right)$	**Essential Discontinuity** $\left(\lim_{x \to a} f(x) \text{ Does Not Exist} \right)$
$f(a)$ **Exists**	 *Point of Displacement (POD)*	 *Jump Discontinuity*
$f(a)$ **Does Not Exist**	 *Point of Exclusion (POE)*	 *Vertical Asymptote*

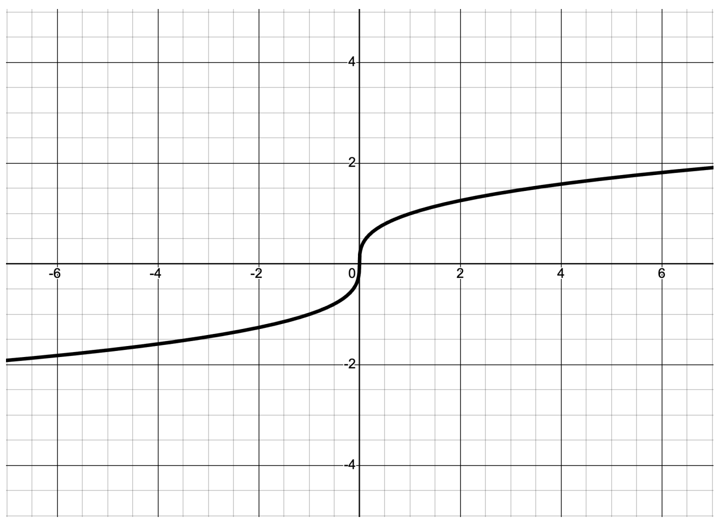
Moving on, let's talk about functions that are not differentiable. "Not differentiable" simply means the derivative does not exist at some point in the function. This can happen one of three ways:

1. A function is not differentiable if it is not continuous (see table above).

2. A function is not differentiable if at some point, the tangent line at that point is vertical.



3. A function is not differentiable if it is not ”smooth”.



The Mean Value Theorem often arises in AP Table Problems like this one:

Ex 4.2.3: Below is a chart showing specific values of the rate of sewage flowing through a pipeline according to time in minutes.

t (minutes)	0	4	6	10	13	15	20
$V(t)$ (gallons/min)	83	68	83	48	38	30	38

Assume $V(t)$ is a continuous and differentiable function. Explain why there are at least two times between $t = 0$ and $t = 20$ when $V'(t) = 0$.

Sol 4.2.3: Because $V(t)$ is a continuous and differentiable function, the Mean Value Theorem applies. Since $V(0) = V(6) = 83$, we have

$$V'(t) = \frac{V(6) - V(0)}{6 - 0} = \frac{0}{6} = 0 \quad \checkmark$$

Similarly, since $V(13) = V(26) = 38$, we have

$$V'(t) = \frac{V(26) - V(13)}{26 - 13} = \frac{0}{13} = 0 \quad \checkmark$$

There may be more c-values where $V'(t) = 0$, but as we can see, we have at least two.

Astute students may notice that we can also apply Rolle's Theorem to this problem, since $f(a) = f(b)$!

Ex 4.2.4: Below is a chart showing specific values of the rate of sewage flowing through a pipeline according to time in minutes.

t (minutes)	0	4	6	10	13	15	20
$V(t)$ (gallons/min)	83	68	83	48	38	30	38

Assume $V(t)$ is a continuous and differentiable function. Explain why there are at least two times between $t = 0$ and $t = 20$ when $V'(t) = 74$.

Sol 4.2.4: For this problem, we have to be a little careful! The Mean Value Theorem would not be very useful in this problem, as we cannot find a $\dfrac{V(b) - V(a)}{b - a}$ that is equal to 74. Instead, let's turn to another theorem in our arsenal: the Intermediate Value Theorem.

Because $V(t)$ is continuous and $68 < 74 < 83$, according to the Intermediate Value Theorem, $V(t)$ will attain the height (or value) of 74 between $V(0) = 83$ and $V(4) = 68$, and $V(4) = 68$ and $V(6) = 83$. Thus, we have found our two times where $V(t) = 74$.

4.2 Free Response Homework

Verify that the following functions fit all the conditions of the Mean Value Theorem, and then find all values of c that satisfy the conclusion of the Mean Value Theorem.

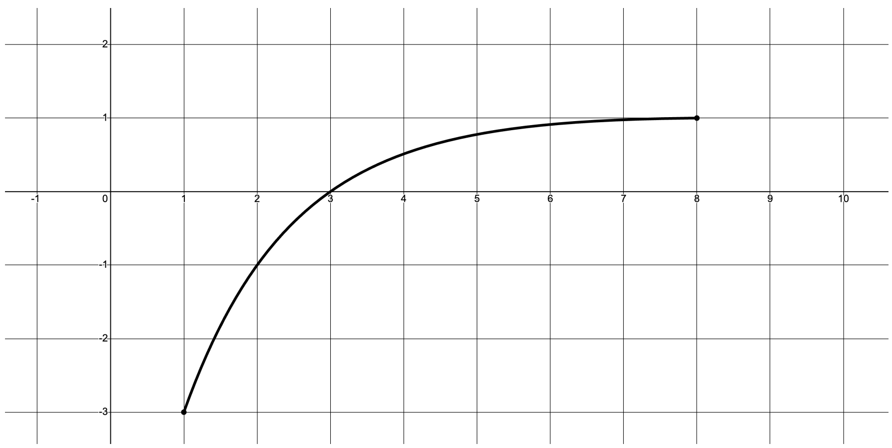
1. $f(x) = x^3 - 3x^2 + 2x + 5 \quad \Big| \quad x \in [0,\ 2]$

2. $f(x) = \sin(2\pi x) \quad \Big| \quad x \in [-1,\ 1]$

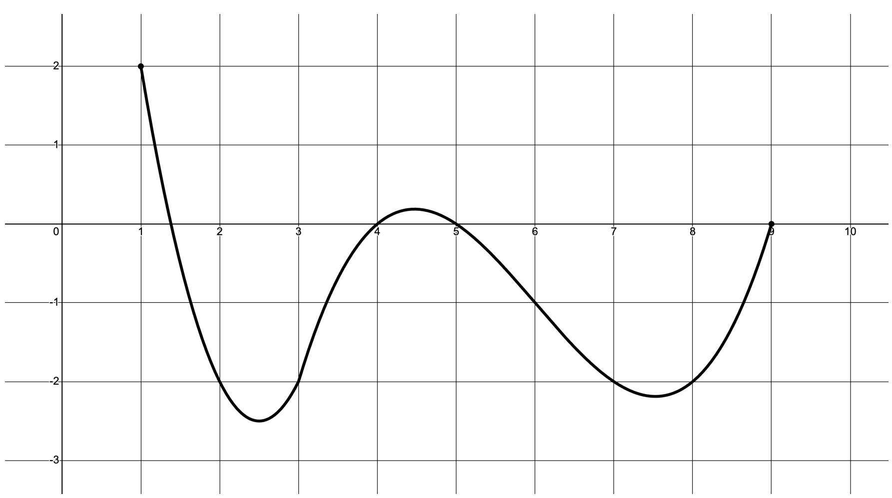
3. $g(t) = t\sqrt{t+6} \quad \Big| \quad t \in [-4,\ 0]$

4. $H(t) = \dfrac{t}{(t-6)^2} \quad \Big| \quad t \in [2,\ 8]$

5. Given the graph of the function below, estimate all values of c that satisfy the conclusion of the Mean Value Theorem for the interval [1, 8].



6. Of the function below, estimate all values of c that satisfy the conclusion of the Mean Value Theorem for the interval [1, 9].



Determine if the following functions fit the conditions of the Intermediate Value Theorem, and determine the range of the results.

7. $f(x) = x^3 - 3x^2 + 2x + 5 \quad \Big| \quad x \in [0,\, 2]$

8. $f(x) = \sin(2\pi x) \quad \Big| \quad x \in [-1,\, 1]$

9. $g(t) = t\sqrt{t+6} \quad \Big| \quad t \in [-4,\, 0]$

10. $H(t) = \dfrac{t}{(t-6)^2} \quad \Big| \quad t \in [2,\, 8]$

11. On May 15th, the weather in the town of Tarcía changes at a rate of $W(t)$ degrees Fahrenheit per hour. $W(t)$ is a twice-differentiable, increasing, and concave up function with selected values in the table below.

t (hours since midnight)	0	1	3	6	8
$W(t)$ (deg Fahrenheit per hour)	-2.4	-2.1	-1.2	1.8	4.5

At midnight ($t = 0$), the weather in Tarcía is 40 degrees Fahrenheit.

(a) Is there a time in $0 \leq t \leq 8$ when the rate of change of the temperature equals 7 degrees Fahrenheit per hour? Justify your answer.

(b) Is there a time in $0 \leq t \leq 8$ when $W(t) = 0$? Justify your answer.

12. Star Formation Rate (SFR) observations of red-shift allow scientists to track the total mass gained in a galaxy by the making of new stars. Below is a table of such data:

t	0	1	2	3	4	5	6	7	8
SFR	0.0029	0.0051	0.0055	0.0049	0.0042	0.0035	0.0029	0.0025	0.0021

SFR is measured in solar masses per cubic parsec per gigayear (millions of years), and t is measured in gigayears.

(a) Is there a time in $0 \leq t \leq 8$ when $SFR = 0$? Justify your answer.

(b) Is there a time in $0 \leq t \leq 8$ when $SFR = 0.0056$? Justify your answer.

4.2 Multiple Choice Homework

1. Let f be a polynomial function with degree greater than 2. If $a \neq b$ and $f(a) = f(b) = 1$, which of the following must be true of at least one value of x between a and b?

I. $f(x) = 0$ II. $f'(x) = 0$ III. $f''(x) = 0$

a) I only b) II only c) III only d) II and III only e) I, II, and III

---

2. Which of the following functions **fail** to meet the conditions of the Mean Value Theorem?

I. $3x^{\frac{2}{3}} - 1$	$x \in [-1, 2]$
II. $\lvert 3x - 2 \rvert$	$x \in [1, 2]$.
III. $4x^3 - 2x + 3$	$x \in [-1, 2]$.

a) I only b) II only c) III only d) I and II only e) II and III only

---

3. $y = -2x^2 + x - 2$ is defined on $x \in [1, 3]$. Find the c-value determined by the Mean Value Theorem.

a) $c = \dfrac{1}{2}$ b) $c = \dfrac{5}{4}$ c) $c = \dfrac{3}{2}$ d) $c = 2$ e) $c = \dfrac{9}{4}$

---

4. Let $g(x)$ be a continuous function on the interval $[0, 1]$ and let $g(1) = 0$ and $g(0) = 1$. Which of the following statements is not necessarily true?

a) There exists a number c on $[0, 1]$ such that $g(c) \geq g(x)$ for all $x \in [0, 1]$.

b) For all a and b in $[0, 1]$, if $a = b$, then $g(a) = g(b)$.

c) There exists a number c on $[0, 1]$ such that $g(c) = \dfrac{1}{2}$.

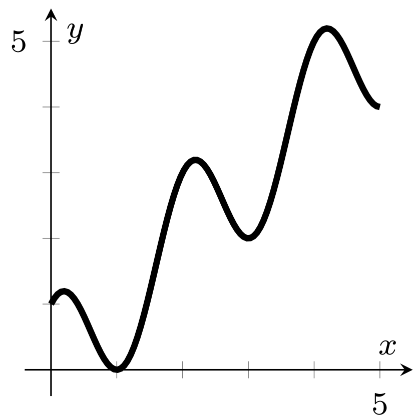
d) There exists a number c on $[0, 1]$ such that $g(c) = \dfrac{3}{2}$.

e) For all c in $(0, 1)$, $\lim\limits_{x \to c} g(x) = g(c)$.

---

5. The graph of a differentiable function f is shown below on the closed interval $[0, 5]$. How many values of x in the open interval $(0, 5)$ satisfy the conclusion of the Mean Value

Theorem for f on $[0,\ 5]$.



a) 1 b) 2 c) 3 d) 4 e) 5

---

6. To which of the following phrases does the equation $Y = \dfrac{f(b) - f(a)}{b - a}$ pertain?

a) The Mean Value Theorem

b) The Average Value Theorem

c) The Intermediate Value Theorem

d) The Extreme Value Theorem

e) Rolle's Theorem

---

7. The table below shows particular values of a differentiable function $E(t)$. At how many times on the interval $t \in [0,\ 12]$ does the Mean Value Theorem guarantee that $E'(t) = 0$?

t (hours)	0	2	4	6	8	10	12	14
$E(t)$	10	4	10	5	7	3	7	5

a) 0 b) 1 c) 2 d) 3 e) 4

4.3: Accumulation of Rates Revisited

In Chapter 3 Section 4, we skipped many questions involving critical and extreme values. We will revisit those problems here in this section.

OBJECTIVES

Solve Critical Values and Extreme Value Problems in the Context of Accumulation of Rates.

Ex 4.3.1:

The Amusement Park Problem (AP 2002)

The rate at which people enter a park is given by the function

$$E(t) = \frac{15600}{t^2 - 24t + 160},$$

and the rate at which they are leaving is given by

$$L(t) = \frac{9890}{t^2 - 38t + 370} - 76.$$

Both $E(t)$ and $L(t)$ are measured in people per hour where t is the number of hours past midnight. The functions are valid for when the park is open, $8 \leq t \leq 24$. At $t = 8$ there are no people in the park.

(d) At what time t, for $8 \leq t \leq 24$, does the model predict the number of people in the park is at at maximum?

(Ex 3.4.1)

Sol 4.3.1:

(d) Recall that $H(t) = \int_8^t (E(x) - L(x))\, dx$. To find the maximum number of people, we have to set the derivative $H'(t)$ equal to zero. **We must also check the endpoints**, which are also critical values themselves.

$$H'(t) = E(t) - L(t) = 0$$

Using a graphing calculator, we find that the zero is at $t = 16.046$. Therefore, our critical values are 8, 16.046, and 24. Now, let's analyze all these values.

1. At $t = 8$, $H(8) = 0$ (this was given at the beginning of the problem)

2. At $t = 16.046$, $H(16.046) = \int_8^{16.046} (E(x) - L(x))\, dx = 5023$

3. At $t = 24$, $H(24) = \int_8^{24} (E(x) - L(x))\,dx = 1453$

Because 5023 is the largest of these three H values, the maximum number of people in the park occurs at $\boxed{t = 16.046}$.

Ex 4.3.2:

The Synthetic Oil Problem

A certain industrial chemical reaction produces synthetic oil at a rate of

$$S(t) = \frac{15t}{1+3t}.$$

At the same time, the oil is removed from the reaction vessel by a skimmer that has a rate of

$$R(t) = 2 + 5\sin\left(\frac{4\pi}{25}t\right).$$

Both functions have units of gallons per hour, and the reaction runs from $t = 0$ to $t = 6$. At time $t = 0$, the reaction vessel contains 2500 gallons of oil.

(d) For the interval indicated above, at what time t is the amount of oil in the reaction vessel at a minimum? What is the minimum value? Justify your answers.

(Ex 3.4.3)

Sol 4.3.2:

(d) Recall that $P(t) = 2500 + \int_0^t (S(x) - R(x))\,dx$. Now, just like the last example, let's set the derivative $P'(t)$ equal to zero and check the endpoints.

$$P'(t) = S(t) - R(t) = 0$$

Using a graphing calculator, we find that the zero is at $t = 5.117$. Analyzing these values, we have:

1. At $t = 0$, $P(0) = 2500$ (Again, this was given in the problem)
2. At $t = 5.117$, $P(5.117) = 2500 + \int_0^{5.117} (S(x) - R(x))\,dx = 2492.367$
3. At $t = 6$, $P(6) = 2500 + \int_0^6 (S(x) - R(x))\,dx = 2493.277$

As 2492.367 is the lowest of these three P values, the minimum amount of oil in the reaction vessel is $\boxed{2492.367 \text{ gallons}}$ and occurs at $\boxed{t = 5.117}$ gallons.

Now, let's do an accumulation of rates problems that combines all of the concepts that we have learned.

Ex 4.3.3:

The Donner Summit Snowfall Problem

The snowfall in Donner Summit is tracked by the US Weather Service. For the month of March in 2022, $S(t)$ represents the rate of snowfall in inches per day and its data is presented in the table below.

t (days)	1	3	4	7	11	15	21
$V(t)$ (inches/day)	1.4	4.9	4.4	0.1	4.6	0.2	2.7

The equation

$$M(t) = 0.65 - 0.35\cos\left(\frac{5x^{0.95}}{6}\right)$$

represents the rate at which the snow melts in inches per day, where t is measured in days.

(a) Find $\int_1^{21} M(t)\,dt$. Using the correct units, explain the meaning of $\frac{1}{21-1}\int_1^{21} M(t)\,dt$ in the context of the problem.

(b) Using a right Riemann Sum, find $\int_1^{21} S(t)\,dt$. State the correct units.

(c) Approximate $S'(5)$. Using the correct units, explain $S'(5)$ in the context of the problem.

(d) Assume

$$S(t) = 2.5 - 2.5\cos\left(\frac{10x^9}{9}\right)$$

would model the snow fall. If there were 9 inches of snow on the ground at the beginning of day 1, find the minimum amount of snow on the ground between $t = 1$ and $t = 7$.

Sol 4.3.3:

(a) Using $\boxed{\texttt{MATH}}$ $\boxed{\texttt{9}}$ on our calculators, we can find that $\int_1^{21} M(t)\,dt = \boxed{13.023 \text{ inches}}$.

Referencing our key phrases and units table, we find that $\frac{1}{21-1}\int_1^{21} M(t)\,dt$ is the **approximate average snow melt, in inches per day, between** $t = 1$ **and** $t = 21$ **days**.

(b)

$$\begin{aligned}\int_1^{21} S(t)\,dt &\approx (3-1)\cdot 4.9 + (4-3)\cdot 4.4 + (7-4)\cdot 0.1 \\ &+ (11-7)\cdot 4.6 + (15-11)\cdot 0.2 + (21-15)\cdot 2.7 \\ &= \boxed{49.9 \text{ inches of snow}}\end{aligned}$$

(c) $S'(5) \approx \frac{6-2}{4-7} = \boxed{-\frac{4}{3}\text{ in/day}^2}$. In the context of the problem, this means that **the rate of snowfall, in inches per day, is decreasing by** $\frac{4}{3}$ **inches per day per day on day 5**.

(d)

$$A(t) = 9 + \int_1^t (S(x) - M(x)),\, dx$$

$$A'(t) = S(t) - M(t) = 0$$

$$t = 0.541,\ 5.751$$

Creating a table to track our values of t and $A(t)$, we have

t	$A(t)$
0	9
0.541	8.781
5.751	22.955
7	22.096

It's clear from the table that the minimum amount of snow on the ground is $\boxed{8.781 \text{ inches}}$.

Make sure to answer the question that is being asked! This question specifically asked for the *amount* of snow.

So, to summarize, to find the extrema of a function, you must first determine the amount equation (the primary function being analyzed) and then identify all critical values. These critical values are found by solving for where the first derivative is zero $\left(\frac{dy}{dx} = 0\right)$, where the derivative does not exist $\left(\frac{dy}{dx} = DNE\right)$, or by identifying the endpoints of any given domain restrictions. Once these values are established, you calculate the corresponding function values for each critical point and organize them into a table for comparison. Finally, use this data to identify the absolute maximums or minimums to provide a specific answer to the original question.

4.3 Free Response Homework

1. The Chlorine Problem

The number of parts per million (ppm), $C(t)$, of chlorine in a pool changes at the rate of

$$C'(t) = 1 - 3e^{-0.2\sqrt{t}}$$

ppm per day, where t is measured in days. There are 50 ppm of chlorine in the pool at time $t = 0$. Chlorine should be added to the pool if the level drops below 40 ppm.

(a) Is the amount of chlorine increasing or decreasing at $t = 9$? Why or why not?

(b) For what value of t is the amount of chlorine at a minimum? Justify your answer.

2. The Ellis Island Problem (II)

In 1920, Dr. Quattrin's grandfather Andrea returned to America from Italy after fighting in World War I. He arrived in New York Harbor on the *SS Pannonia* and, despite having established residency in 1913, had to be processed through the Immigration Center at Ellis Island. There were 1123 non-citizen, third-class passengers on the *Pannonia* that had to go through processing. Immigrants entered the processing line at a rate modeled by the function

$$E(t) = 8843\left(\frac{t}{5}\right)^4\left(1 - \frac{t}{10}\right)^5.$$

where t, in $0 \leq t \leq 10$, is measured in hours after the ship began offloading immigrants. The new arrivals were processed out a rate of 250 people per hour. The *Pannonia* was the third ship in port, so there were already 2500 people in line when the *Pannonia* passenger got into line.

Assuming no new ships entered the harbor, what was the absolute maximum number of people in the processing line? Justify your answer.

3. The Post Office Letter Problem (II)

Letters arrive at a post office at a rate of

$$P(t) = 8 + t\sin\left(\frac{t^3}{80}\right)$$

hundred letters per hour over the course of a workday. The day begins at 9 am ($t = 0$) and ends at 5 pm ($t = 8$). There are three hundred letters in the office at 9 am. Workers send letters out of the office at a constant rate of five hundred letters per hour.

(a) Write an expression for $L(t)$, the total number of letters in the post office at time t.

(b) What is the maximum number of letters in the office over the course of the workday $(0 \leq t \leq 8)$?

4. The Ski Resort Problem

A ski resort uses a snow machine to control the snow level on a ski slope. Over a 24-hour period the volume of snow added to the slope per hour is modeled by the equation

$$S(t) = 24 - t\sin^2\left(\frac{t}{14}\right).$$

The rate that the snow melts is modeled by

$$M(t) = 10 + 8\cos\left(\frac{t}{3}\right).$$

Both $M(t)$ and $S(t)$ are measured in yd^3/h and t is measured in hours for $0 \leq t \leq 24$. At time $t = 0$, the slope holds 50 yd^3 of snow.

(a) Write an expression for $A(t)$, the total amount of snow on the mountain at time t.

(b) Find the absolute maximum and minimum amount of snow on the mountain during $0 \leq t \leq 24$. Indicate the units.

5. The Trinary Star Problem (II)

More than 30% of observed star systems have multiple stars, and 70% of those have more than two stars. When stars are closed together, they exchange mass in a process known as accretion. Consider a trinary system where S_1 is larger than S_2, and S_2 is larger than S_3. S_3 will lose mass to S_2, and S_2 will lose mass to S_1. While scientific readings are not available because of the time scale, let us suppose that S_2 loses mass to the larger S_1 at a rate of

$$L(t) = 1 + (0.01t)^2 + 0.23\sin\left(\frac{\pi}{25}t\right)$$

and gains mass from the smaller S_3 at a rate of

$$G(t) = 0.2 + 0.15\sqrt{t}$$

where $0 \leq t \leq 100$ years. $L(t)$ and $G(t)$ are measured in yottatons per year (Y/yr). (A yottaton is 10^{26} tons, or 10^{-7} solar masses.)

(a) Write an expression for $M(t)$, the total mass of S_2 at time t.

(b) If the mass of S_2 is $1,000,000$ yottatons at $t = 0$, find the minimum mass of S_2 on the time interval $0 \leq t \leq 100$.

6. The Flooded Basement Problem (II)

The basement of a house is flooded, and water keeps pouring in at a rate of

$$w(t) = 95\sqrt{t}\sin^2\left(\frac{t}{6}\right)$$

gallons per hour. At the same time, water is being pumped out at a rate of

$$r(t) = 275\sin^2\left(\frac{t}{3}\right).$$

When the pump is started, at time $t = 0$, there is 1200 gallons of water in the basement. Water continues to pour in and be pumped out for the interval $0 \leq t \leq 18$.

(a) Write an expression for $G(t)$, the total gallons of water in the basement at time t.

(b) Find the minimum amount of water in the basement on the time interval $0 \leq t \leq 18$.

7. The King Philip Problem

On Ocean Beach at the foot of Noriega St. is a shipwreck buried in the sand. In 1858, the clipper King Philip ran aground. It was stripped of salvageable material, but the 45% of its hull is still buried on the beach, and it appears every few years as the tide washes the sand in and out. As the sand washes in or is added by the National Park Service, the height changes at a rate of

$$A(t) = \frac{6t}{1+2t},$$

and, as the tide washes sand out, the height changes at a rate of

$$B(t) = 2 + 7\cos\left(\frac{\pi}{15}t\right)\sin\left(\frac{3\pi}{16}t\right)$$

Both $A(t)$ and $B(t)$ are both measured in inches above the wreck per year for $0 \leq t \leq 10$. At $t = 0$, the height is 10 inches above the wreck.

(a) Find the total inches of sand above the wreck which are washed out in the first five years. Indicate the correct units.

(b) Is the rate of change of the height increasing or decreasing at $t = 6$ years? Justify your answer.

(c) Write an expression for $H(t)$, the total number of inches of sand above the wreck at time t.

(d) What is the absolute minimum number of inches of sand above the wreck over the course of the ten years described in the problem?

8. The Cat Population Problem (II)

The Peninsula Humane Society (PHS) is dedicated to the care and adoption of as many animals who they receive as possible. Since cats breed seasonally, the number of cats and kittens they receive into their facility in a given year varies roughly sinusoidally with time. The data available from 2019 show the rate $R(t)$, measured in healthy cats per month, varies with time, measured in months after New Years Day, according to the equation

$$R(t) = 120 - 88\cos\left(\frac{\pi}{6}(t-2)\right).$$

The rate $A(t)$ at which adoption occur, measured in cats per month, varies with time, measured in months after New Year's Day, according to the equation

$$A(t) = 125 - 85\cos\left(\frac{\pi}{6}(t-3)\right).$$

On New Years' Day ($t = 0$), there were 131 cats in the PHS nursery waiting to be adopted.

(a) How many cats and kittens were received at PHS in 2019?

(b) Find $A'(10.3)$. Using the correct units, explain the meaning of $A'(10.3)$ in the context of the problem.

(c) Find the number of healthy cats and kittens predicted by the models to be in the PHS facility at the end of 2019.

(d) Find the time when the number of healthy cats and kittens in the PHS facility during 2019 was at an absolute maximum. Indicate the units.

9. The Oak Island Problem

The search for treasure on Oak Island has been ongoing since 1795. In the last ten years, modern engineering techniques have been brought to bear by the Lagina brothers. They hired ROC Drilling to sink 8-foot-diameter metal tubes (called caissons) down to 200 feet and then use a hammer grab to excavate the "spoils." The hammer grab brings up the spoils at a rate of

$$H(t) = 9 + 4\cos\left(\frac{t^2}{14}\right).$$

The spoils are then taken to a wash plant where they are rinsed and sorted into piles by size so they can be visually inspected to find artifacts. A shift supervisor measures their rate of output every few hours and records the findings in the chart below.

t (hours)	0	1	3	5	8
$P(t)$ (yd^3/hour)	0	8.1	14.5	12.3	3.2

At the beginning of the day there are 2 yd^3 at the wash plant from the day before.

(a) Find the total amount of spoils pulled from the caisson by the hammer grab during the eight hours described on the table. Indicate the correct units.

(b) Use a right Riemann Sum with sub-intervals indicated by the table to approximate $\int_0^8 P(t)\,dt$ Using correct units, explain the meaning of this value in the context of the problem.

(c) Assume the equation

$$W(t) = 0.1t(10-t)^2$$

models the $P(t)$ data in the table. Write an expression for $U(t)$, the amount of unwashed spoils at time t.

(d) What is the absolute maximum amount of unwashed spoils waiting to be cleaned?

10. The Callaghan Ranch Harvest Problem

In 1867, Dr. Quattrin's great-great grandfather Michael Callaghan bought 160 acres of public land in Solano County to raise wheat and sheep. By 1890, the ranch had grown to 1210 acres. Prior to the advent of the combine harvester, reaping and threshing were done by different machines. The table below shows the rate at which wheat on the ranch was gathered by a *McCormick* Reaper on a given 12-hour workday.

t (hours)	0	2	5	9	12
$R(t)$ (acres/hour)	3.1	2.5	1.6	2.7	1.6

Time t is measured in hours after 6 am ($t = 0$) and $R(t)$ is measured in acres per hour. The wheat was then delivered to the barn and put through the threshing machine. At the start of the day, there are 2.4 acres of wheat in the barn.

(a) Is the rate at which the wheat is being harvested at 9 : 30 am ($t = 3.5$) increasing or decreasing? Explain the result in the context of the problem using the appropriate units.

(b) Use a right-hand Riemann approximation to approximate the total amount of wheat harvested on this particular day.

(c) The thresher processed the wheat into grain and hay at a rate of

$$P(t) = \frac{2.1x}{\sqrt{x^2+1}}$$

acres per hour. How much wheat has been threshed by noon ($t = 6$) when the farm hands break for lunch?

(d) Write an equation for $0 \le t \le 12$ which would determine the amount of wheat in the barn at any time t. Using your estimate for the amount of wheat reaped in part (b) above, find the amount of unthreshed wheat in the barn at the end of the day.

(e) Assume $R(t)$ can be modeled by the equation

$$Q(t) = 3.3 - \frac{3}{20}t - 0.9\sin\left(\frac{\pi}{7}t\right)$$

What is the minimum amount of unthreshed wheat in the barn?

11. The Yuma Desalting Problem

The desalting plant at Yuma, AZ removes alkaline (salt) products from the Colorado River to make the water better for irrigation downstream in Mexico. Data from a pilot run of the plant shows that water enters the plant at a rate $W(t)$ as shown on the table below:

t (months)	0	1	2	3	4	5
$W(t)$ (acre-feet/month)	0	2375	3189	3411	3207	2169
t (months)	6	7	8	9	10	
$W(t)$ (acre-feet/month)	2269	2151	2167	3022	2293	

The rate $P(t)$ of outflow of processed water, in acre-feet per month, is modeled by

$$P(t) = -0.55t^4 + 15t^3 - 158t^2 + 722t + 1032$$

for $0 \le t \le 10$. Based on supplies available, not all the water gets processed before returning to the Colorado River.

(a) Using a midpoint Riemann Sum, approximate the volume of water that enters the plant during these ten months.

(b) Set up an equation for $U(t)$ which would define the amount of unprocessed water that exits the plant. Using your answer in part (a), approximate $U(10)$. Indicate the units.

(c) Approximate $W'(6)$. Using the correct units, explain the meaning of your answer in the context of the problem.

(d) Assuming $W(t)$ can be modeled by

$$E(t) = 2800 + 750 \sin\left(\frac{2\pi}{11}t\right),$$

find the time at which there is an absolute maximum amount of unprocessed water flowing through the plant for $0 \leq t \leq 10$. Justify your answer.

12. The 49er Sack Leader Problem (II)

The table below shows Nick Bosa's sack rate, in sacks per game, over his first four seasons.

g (games completed)	0	19	21	41	57
$B(g)$ (sacks per game)	0	0.68	0.62	0.98	1.16

(a) Using a right-hand Riemann Approximation, determine the approximate number of sacks Nick Bosa had during these four years. Round to the nearest whole number.

(b) Using the data on the table, estimate $B'(32)$. Based on this estimate, was Bosa's sack total increasing at an increasing or decreasing rate? Using the correct units, explain your answer.

(c) Disregarding his injury-shortened 2020 season, a model for Bosa's sack rate per game can be given with the equation

$$S_B(g) = 0.15\sqrt{g}.$$

Find $\frac{1}{57}\int_0^{57} S_B(g)\,dg$. Using the correct units, explain this result in the context of the problem.

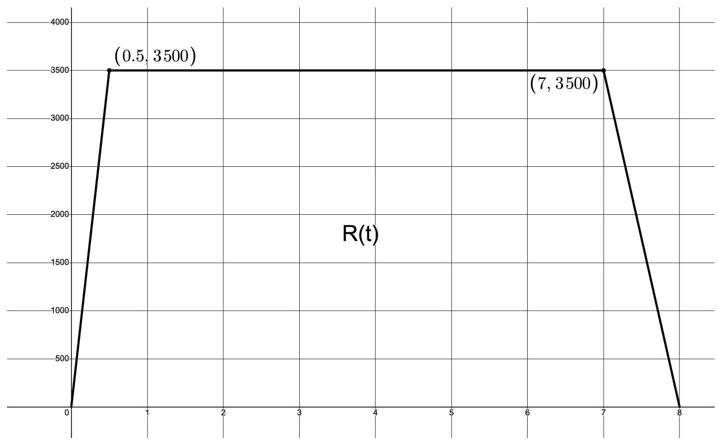
(d) Assume that

$$A(g) = -0.001g^2 + 0.073g - 0.004$$

models the rate of Aldon Smith's sacks over his first 50 games from 2011 to 2014. During what game is the difference between Bosa's and Smith's sack totals at a maximum? Show your calculations.

13. The Groundwater Replenishment System Problem

Disparagingly referred to as *Toilet to Tap*, Orange County's GWRS (Groundwater Replenishment System) has converted 400 billion gallons of raw sewage into drinkable water over the past fifteen years. That is about $25,000$ gallons every eight hours. The process occurs in three phases: microfiltration, reverse osmosis, and ultraviolet disinfection. Let us assume that the rate $R(t)$, in gallons per hour, at which raw sewage enters the process is modeled by the graph below, formed by three linear functions.



Further, let use assume the rate at which the raw sewage is converted to treated sewage, in gallons per hour, in the microfiltration process is modeled by

$$T(t) = 83e^{0.5t}\sqrt{8t - t^2}$$

for $t \in [0,\, 8]$ hours.

(a) How many gallons of raw sewage have entered the process in these eight hours?

(b) Find $T'(6.2)$. Using the correct units, explain the meaning of this result in the context of the problem.

(c) Assuming there is no raw sewage left from the previous process, write an expression for $A(t)$, the total amount of sewage present within the process at any time t.

(d) Find the maximum amount of raw sewage in process during the microfiltration phase.

(e) Was all the raw sewage converted? Explain your reasoning.

14. The Groundwater Replenishment System Problem (II)

Disparagingly referred to as *Toilet to Tap*, Orange County's GWRS (Groundwater Replenishment System) has converted 400 billion gallons of raw sewage into drinkable water over the past fifteen years. That is about $25,000$ gallons every eight hours. The process occurs in three phases: microfiltration, reverse osmosis, and ultraviolet disinfection. Let us assume that the rate $T(t)$, in gallons per hour, at which treated sewage enters the reverse osmosis process is modeled by

$$T(t) = 3800e^{-0.4t}\sqrt{8t - t^2}$$

gallons for $0 \le t \le 8$ hours.

Further, let us assume that the rate $N(t)$ at which the treated sewage is converted to non-potable (non-drinkable) water, in gallons per hour, in the reverse osmosis process is modeled by the table below.

t (hours)	0	1.8	4.4	6.3	8
$N(t)$ (gallons per hour)	0	2301	3107	4320	5204

(a) How many gallons of treated sewage have entered the process in these eight hours? Round to the nearest gallon.

(b) Find $N'(5.2)$. Using the correct units, explain the meaning of the result in the context of the problem.

(c) Use a trapezoidal sum to approximate the amount of non-potable water that exits the system in these eight hours. Round to the nearest gallon.

(d) Assuming these is 1200 gallons of treated sewage in the system at the left from the previous process, use the answers to part (a) and (c) above to approximate, to the nearest gallon, the amount of treated sewage still in the system at the end of these eight hours.

(e) Assume

$$M(t) = -1 + 1650\sqrt{t} + 352\cos\left(\frac{\pi}{4}x\right)$$

is a model of the $N(t)$ data given above. What is the maximum amount, to the nearest gallon, of water in the system for $0 \le t \le 8$ hours? Justify your answer.

15. The Groundwater Replenishment System Problem (III)

Disparagingly referred to as *Toilet to Tap*, Orange County's GWRS (Groundwater Replenishment System) has converted 400 billion gallons of raw sewage into drinkable water over the past fifteen years. That is about $25,000$ gallons every eight hours. The process occurs in three phases: microfiltration, reverse osmosis, and ultraviolet disinfection. Let us assume that $N(t)$ and $P(t)$, in gallons per hour, are the rates at which non-potable (non-drinkable) water enters and the potable (drinkable) water exits the ultraviolet disinfection process. Further, let us assume that $N(t)$ and $P(t)$ are modeled by the equations

$$N(t) = 3750 - 3000\cos\left(\frac{\pi}{8}x^{1.5}\right)$$

and

$$P(t) = 1125\sqrt{8x - x^2}$$

for $t \in [0, 8]$ hours.

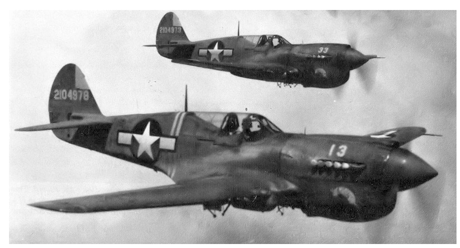
(a) How many gallons of non-potable water have entered the process in these eight hours?

(b) Find $P'(5.3)$. Using the correct units, explain the meaning of the result in terms of the situation.

(c) Assuming there are 4500 gallons of non-potable water left from the previous process, write an expression for the $A(t)$, the total amount of non-potable water present within the process at any time t.

(d) Find the time at which the minimum amount of non-potable water in process during the microfiltration phase.

16. The WWII Aircraft Problem



The table below shows the rates, in thousand of military planes per month, at which aircraft were built during World War II, where t is measured in months after the beginning of 1939. $A(t)$ represents the rate at which the Axis powers (Germany and Japan) were building aircraft, and $U(t)$ represents the rate at which the United States was building aircraft. During the first six months of the war, the Axis powers had produced 4.9 thousand planes and the US had produced 1.2 thousand planes.

t (months)	6	18	30	42	54	66
$A(t)$ (thousand planes per month)	1.06	1.3	1.4	2.1	3.5	5.7
$U(t)$ (thousand planes per month)	0.18	0.5	1.6	4.0	7.2	8.1

(a) Using a right-hand Riemann Sum, approximate the number of aircraft built by the Axis powers during these 66 months

(b) Approximate $U'(20)$. Using the correct units, explain the meaning of this result

(c) Assume

$$X(t) = 0.757e^{0.028t}$$

models the data for $A(t)$ and

$$S(t) = 0.164e^{0.067t}$$

models the data for $U(t)$. Set up an equation that would model the difference between the Axis and US aircraft production.

(d) Find the maximum difference between the Axis and US aircraft production

4.4: The Second Derivative: Points of Inflection and Concavity

In this section, we are going to take a look at the graphical applications of the second derivative. But first, let's review some definitions:

Concavity $\rightarrow$ The direction a curve bends.

Concave Up $\rightarrow$ The curve bends counter-clockwise (upwards like a cup)

Concave Down $\rightarrow$ The curve bends clockwise (downwards like a frown)

Inflection Value $\rightarrow$ The x-value at which the concavity of the curve switches. (Note: this can be a vertical asymptote.)

Point of Inflection $\rightarrow$ The point at which the concavity of the curve switches.

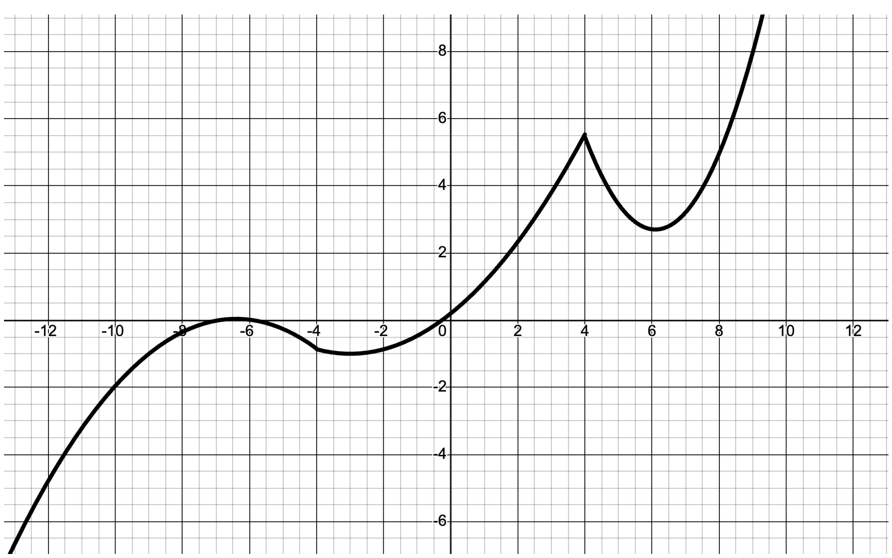
Just as we use the first derivative to find extrema, we use the second derivative to find points of inflection. Therefore, we can say that:

A point of inflection (POI) occurs if and only if the sign of $f''(x)$ changes.

OBJECTIVES

Find Points of Inflection and Intervals of Concavity.

Before we attempt any examples, let's take a look at a basic graphical representation of concavity. Take this graph for example:



It appears that this graph is concave down on $(-\infty, 4)$ and concave up on $(-4, 4) \cup (4, \infty)$, more or less. Therefore, $x = -4$ would be an inflection value. Note that $x = 4$ is not an inflection value because the concavity of the graph does not change at that value.

Ex 4.4.1: Find the point(s) of inflection of $y = x^3 + 4x^2 - 3x + 1$.

Sol 4.4.1: To find the point(s) of inflection of this function, we need to know the sign(s) of the second derivative. We can begin by finding just that.

$$y' = 3x^2 + 8x - 3$$

$$y'' = 6x + 8$$

Setting the second derivative equal to zero and making a sign pattern gives us:

$$6x + 8 = 0 \implies x = -\frac{4}{3}$$

$$\begin{array}{cccc} y'' & - & 0 & + \\ & \longleftarrow & \!\!\!\longrightarrow & \\ x & & -\frac{4}{3} & \end{array}$$

So, since the sign of the second derivative changes at $x = -\frac{4}{3}$, we know that $x = -\frac{4}{3}$ is an inflection value. To find the point of inflection, we simply plug this value back into our original equation to obtain our final answer of $\boxed{\left(-\frac{4}{3}, 5.899\right)}$.

Ex 4.4.2: Find the point(s) of inflection of $y = \dfrac{4x}{x^2+1}$.

Sol 4.4.2: Once again, let's take the second derivative, set it equal to zero, and create a sign pattern.

$$\begin{aligned} y' &= \frac{(x^2+1)(4) - (4x)(2x)}{(x^2+1)^2} \\ &= \frac{-4x^2+4}{(x^2+1)^2} \end{aligned}$$

$$y'' = \frac{\left(x^2+1\right)^2(-8x) - 2\left(-4x^2+4\right)\left(x^2+1\right)(2x)}{\left(x^2+1\right)^4}$$

$$= \frac{8x\left(x^2+1\right)\left(-\left(x^2+1\right) - \left(-2x^2+2\right)\right)}{\left(x^2+1\right)^4}$$

$$= \frac{8x\left(x^2-3\right)}{\left(x^2+1\right)^3}$$

Setting the second derivative equal to zero and making a sign pattern gives us:

$$\frac{8x\left(x^2-3\right)}{\left(x^2+1\right)^3} = 0 \implies x = 0, \pm\sqrt{3}$$

y''	$-$	0	$+$	0	$-$	0	$+$
x		$-\sqrt{3}$		0		$\sqrt{3}$	

Because the sign of the second derivative changes at all of our x-values, we know that all of our x values are points of inflections. Plugging these values back into our original equation gives us our final answer of $\boxed{\left(-\sqrt{3}, -\sqrt{3}\right), (0, 0), \left(\sqrt{3}, \sqrt{3}\right)}$.

Ex 4.4.3: Given the following two sign pattens:

$\frac{dy}{dx}$	$-$	0	$+$	0	$-$	0	$-$	0	$+$
x		-9		-3		-1		7	

$\frac{d^2y}{dx^2}$	$+$	0	$-$	0	$+$	0	$-$	0	$+$
x		-7		-2		0		5	

(a) On what interval(s) is y decreasing?

(b) On what interval(s) is y concave up?

(c) On what interval(s) is y both increasing and concave down?

Sol 4.4.3:
(a) To find the intervals on which y is decreasing, we look for the intervals where the

first derivative of y is negative. From our sign pattern, we see that our desired interval is $\boxed{x \in (-\infty, -9) \cup (-3, 1) \cup (-1, 7)}$.

(b) To find the intervals on which y is concave up, we look for the intervals where the second derivative of y is positive. From our sign pattern, we see that our desired interval is $\boxed{x \in (-\infty, -7) \cup (-2, 0) \cup (5, \infty)}$.

(c) To find the intervals on which y is both increasing *and* concave down, we need to take a look at each interval separately and find the intersection (or what's included in both) of the two intervals. From our sign pattern, we know that y is increasing on the interval $x \in (-9, -3) \cup (7, \infty)$. We also see that y is concave down on the interval $x \in (-7, -2) \cup (0, 5)$. Taking the intersection of these two intervals gives us our desired answer of $\boxed{x \in (-7, -3)}$.

Ex 4.4.4: Find the intervals of concavity for $y = xe^{2x}$.

Sol 4.4.4: We begin this problem just like with the others:

$$\begin{aligned} y' &= x(2)\left(e^{2x}\right) + e^{2x}(1) \\ &= e^{2x}(2x+1) \end{aligned}$$

$$\begin{aligned} y'' &= e^{2x}(2) + (2x+1)(2)\left(e^{2x}\right) \\ &= e^{2x}(2 + (2x+1)(2)) \\ &= e^{2x}(4x+4) \end{aligned}$$

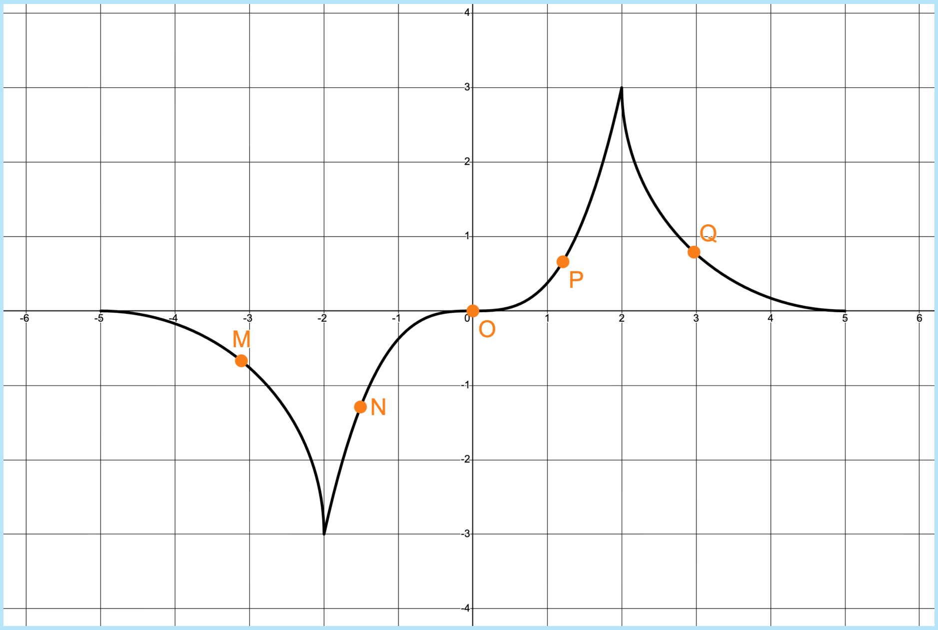
Now, let's set the second derivative equal to zero and create a sign pattern:

$$e^{2x}(4x+4) = 0 \implies x = -1$$

$$\begin{array}{cccc} y'' & - & 0 & + \\ & \longleftarrow & & \longrightarrow \\ x & & -1 & \end{array}$$

From this sign pattern, we conclude that y is $\boxed{\text{concave down on } x \in (-\infty, -1)}$ and $\boxed{\text{concave up on } x \in (-1, \infty)}$.

Ex 4.4.5: At what point on the curve below is $\dfrac{dy}{dx} < 0$ and $\dfrac{d^2y}{dx^2} > 0$?



a) M b) N c) O d) P e) Q

Sol 4.4.5: This problem is very similar to **Ex 4.4.3**, but instead of sign patterns, we are given a graph. Now, let's break down the information we are given:

First, $\frac{dy}{dx} < 0$ implies that the graph is decreasing at our desired point. Already, that information allows us to rule out points N, O, and P. The second piece of information that we are given is $\frac{d^2y}{dx^2} > 0$. This means that the curve is concave up at our desired point. From our remaining two points, we see that point $\boxed{Q}$ satisfies this statement indeed.

4.4 Free Response Homework

1. Given this sign pattern for the second derivative of $F(x)$, on what intervals is $F(x)$ concave down?

$F''(x)$	$-$	0	$+$	0	$-$	0	$+$
x		-3		0		3	

2. The sign pattern for the derivative of $H(x)$ is given.

$\dfrac{d^2H}{dx^2}$	$+$	0	$-$	0	$-$	0	$+$
x		-4		-1		2	

(a) Is $x = -4$ point of inflection?

(b) Is $x = -1$ a point of inflection?

Find point(s) of inflection and interval(s) of concavity for the following equations.

3. $y = x^3 + x^2 - 7x - 15$

4. $y = 3x^4 - 20x^3 + 42x^2 - 26x + 16$

5. $y = -\dfrac{4x}{x^2+4}$

6. $y = \dfrac{x^2-1}{x^2-4}$

7. $y = \ln\left(4x - x^3\right)$

8. $y = x\sqrt{8-x^2}$

9. $y = \dfrac{1}{2}x + \sin(x) \quad \Big| \quad x \in (0,\, 2\pi)$

10. $y = xe^{-x}$

11. $y = e^{-x^2}$

12. $y = \dfrac{x}{x^2-9}$

13. $y = 2x - x^{\frac{2}{3}}$

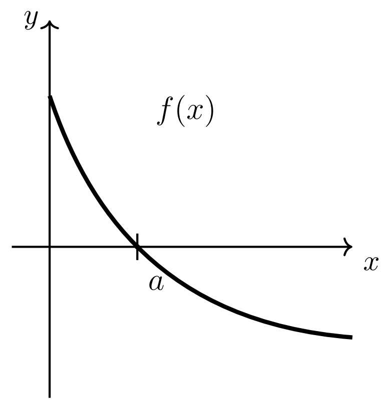
4.4 Multiple Choice Homework

1. $f(x) = x^3 - 6x^2$ is concave up when:

a) $x > 2$ b) $x < 2$ c) $0 < x < 4$ d) $x < 0$ or $x > 4$ e) $x > 6$

---

2. Given the twice differentiable function f, which of the following statements is true?



a) $f'(a) < f''(a) < f(a)$ b) $f'(a) < f(a) < f''(a)$ c) $f''(a) < f(a) < f'(a)$
d) $f''(a) < f'(a) < f(a)$ e) $f(a) < f'(a) < f''(a)$

---

3. For $x > 0$, let $f'(x) = \dfrac{\ln x}{x}$ and $f''(x) = \dfrac{1 - \ln x}{x^2}$. Which of the following is true?

a) f is decreasing for $x > 1$ and the graph of f is concave down for $x > e$.

b) f is decreasing for $x > 1$ and the graph of f is concave up for $x > e$.

c) f is increasing for $x > 1$ and the graph of f is concave down for $x > e$.

d) f is increasing for $x > 1$ and the graph of f is concave up for $x > e$.

e) f is increasing for $0 < x < e$ and is concave down for $0 < x < e^{\frac{3}{2}}$.

---

4. Consider the function $f(x) = \left(x^2 - 5\right)^3$ for all real numbers x. The number of inflection points for the graph of f is

a) 1 b) 2 c) 3 d) 4 e) 5

---

5. If $\frac{d}{dx}[f(x)] = g(x)$ and $\frac{d}{dx}[g(x)] = f(3x)$, then $\frac{d^2}{dx^2}\left[f\left(x^2\right)\right]$ is

a) $4x^2 f\left(3x^2\right) + 2g\left(x^2\right)$ b) $f\left(3x^2\right)$ c) $f\left(x^4\right)$

d) $2xf\left(3x^2\right) + 2g\left(x^2\right)$ e) $2xf\left(3x^2\right)$

---

6. If $y = e^{kx}$, then $\frac{d^5y}{dx^5} =$

a) k^5e^k b) k^5e^{kx} c) $5!e^{kx}$ d) $5!e^x$ e) $5e^{kx}$

---

7. The graph of $y = 3x^5 - 10x^4$ has an inflection point at

a) (0, 0) only b) (3, −81) only c) (2, −6) only

d) (0, 0) and (2, −64) e) (0, 0) and (3, −81)

---

8. An equation of the line tangent to the graph of $y = x^3 + 3x^2 + 2$ at its point of inflection is

a) $y = -3x + 1$ b) $y = -3x = 7$ c) $y = x + 5$

d) $y = 3x + 1$ e) $y = 3x + 7$

---

9. The number of inflection points for the graph of $y = 2x + \cos\left(x^2\right)$ in the interval $0 \leq x \leq 5$ is

a) 6 b) 7 c) 8 d) 9 e) 10

---

10. On which of the following intervals is the graph of the curve $y = x^5 - 5x^4 + 10x + 15$

concave up?

I. $x < 0$ II. $0 < x < 3$ III. $x > 3$

a) I only b) II only c) III only d) I and II only e) II and III only

---

11. Let f be a function with a second derivative given by $f''(x) = x^2(x-3)(x-6)$. What are the x-coordinates of the points of inflection of the graph of f?

a) 0 only b) 3 only c) 0 and 6 only d) 3 and 6 only e) 0, 3, and 6

---

12. The derivative of a function is given by $f'(x) = x^2 \cos\left(x^2\right)$. How many points of inflection does the graph of f have on the open interval $(-2, 2)$?

a) 1 b) 2 c) 3 d) 4 e) 5

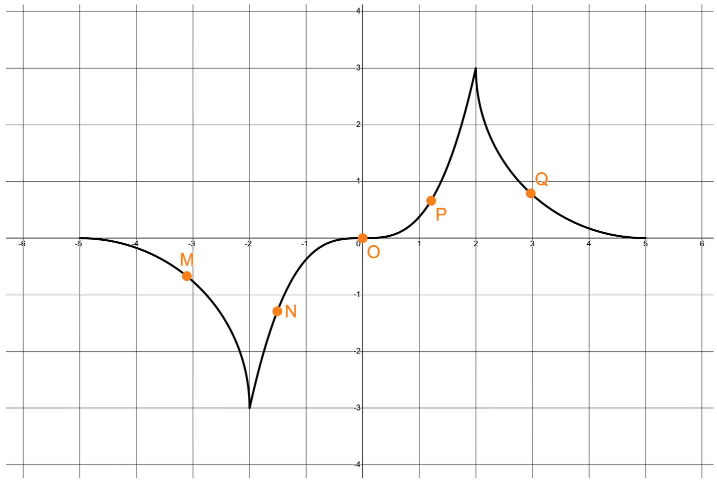
---

13. How many points of inflection does the graph of $y = \cos(x) + \frac{1}{3}\cos(x) - \frac{1}{5}\cos(5x)$ have on the interval $0 \le x \le \pi$?

a) 1 b) 2 c) 3 d) 4 e) 5

---

14. At what point on the curve below is $\frac{dy}{dx} > 0$ and $\frac{d^2y}{dx^2} > 0$?



a) M b) N c) O d) P e) Q

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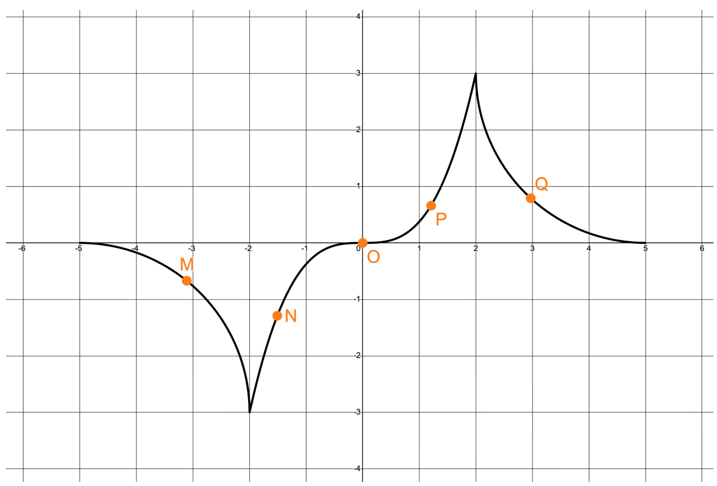
15. At what point on the curve below is $\dfrac{dy}{dx} < 0$ and $\dfrac{d^2y}{dx^2} < 0$?



a) M b) N c) O d) P e) Q

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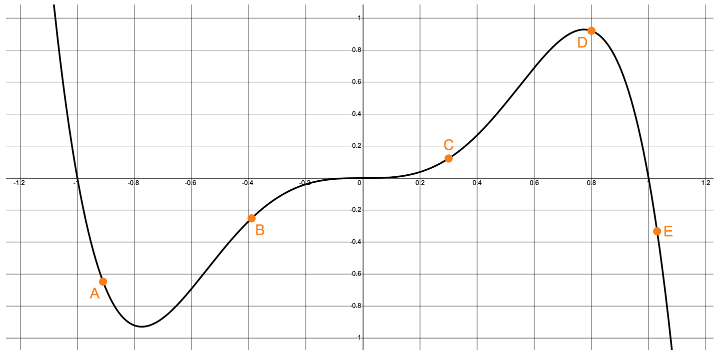
16. The graph of the function f(x) is shown below. At which point on the graph of $f(x)$ is $f'(x) > 0$ and $f''(x) < 0$?



a) A b) B c) C d) D e) E

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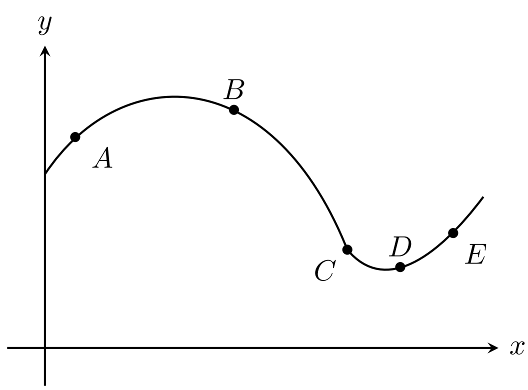
17. At which of the five points on the graph in the figure below is $\dfrac{dy}{dx}$ and $\dfrac{d^2y}{dx^2}$ both negative?



a) A b) B c) C d) D e) E

4.5: Graphing with Derivatives

All last year, we concerned ourselves with sketching graphs based on traits of a function. We tended to look at the one key aspect of the derivative – that is, finding extremes – as it applied to a function expressed by an equation.

Towards the end of the year in precalculus, we began to look at the first and second derivatives as the more important traits of the function and used sign patterns to determine how the points were connected. We were still viewing the function as expressed by an equation, but the emphasis became how the points were connected, instead of the points themselves. We were able to get this information from the sign patterns.

Now, in this section, we are going to put all that information together to sketch more accurate graphs.

OBJECTIVES

Sketch the Graph of a Function Using Information from its First and/or Second Derivatives.
Sketch the Graph of a First and/or Second Derivative from the Graph of a Function.

Before we begin, though, let's quickly summarize all the graphical features that we've learned.

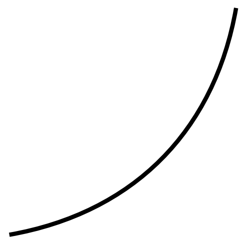
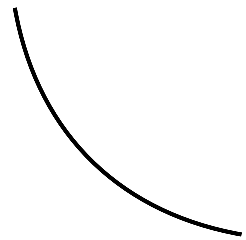
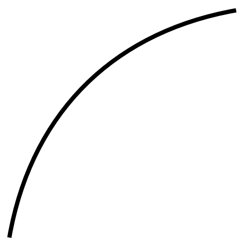
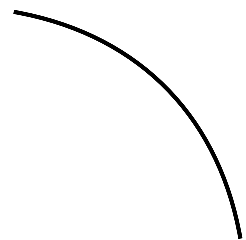
Key Traits

⋆ Domain	⋆ Range	⋆ y-intercept	⋆ Zeroes
⋆ Vertical Asymptotes	⋆ Points of Exclusion	⋆ End Behavior	⋆ Extreme Points

Interpretation of Sign Patterns

Sign	$+$	0 or dne	$-$
$f(x)$	Curve above x-axis	Zero / VA	Curve below x-axis
$f'(x)$	Increasing	Critical Value	Decreasing
$f''(x)$	Concave Up	POI	Concave Down

Graphical 1st, 2nd Derivative Relationships

	Increasing	Decreasing
Concave Up		
Concave Down		

Ex 4.5.1: Make a sign pattern table and create a sketch of $y = x^4 - 2x^3$.

Sol 4.5.1: First, let's create our three sign patterns for the zeroes, extreme values, and points of inflection of this function.
Zeroes:

$$x^4 - 2x^3 = 0 \implies x = 0,\, 2 \qquad \to (0,\, 0),\, (2,\, 0)$$

$$\begin{array}{cccccc} y & + & 0 & - & 0 & + \\ & \longleftarrow & & & & \longrightarrow \\ x & & 0 & & 2 & \end{array}$$

Extreme Values:

$$\frac{dy}{dx} = 4x^3 - 6x^2 = 0 \implies x = 0,\, \frac{3}{2} \qquad \to (0,\, 0),\, \left(\frac{3}{2},\, -\frac{27}{16}\right)$$

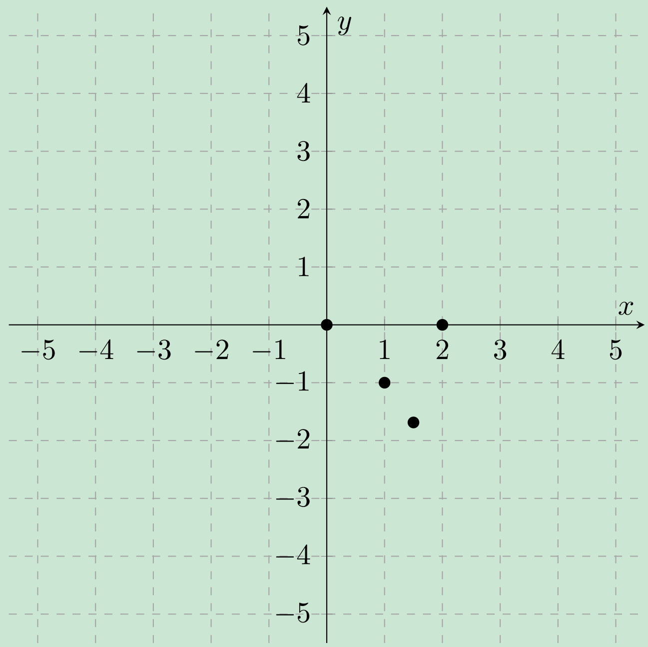
$$\begin{array}{cccccc} y' & - & 0 & - & 0 & + \\ & \longleftarrow & & & & \longrightarrow \\ x & & 0 & & \frac{3}{2} & \end{array}$$

Point of Inflection:

$$\frac{d^2y}{dx^2} = 12x^2 - 12x = 0 \implies x = 0,\, 1 \qquad \to (0,\, 0),\, (1,\, -1)$$

y'	$+$	0	$-$	0	$+$
x		0		1	

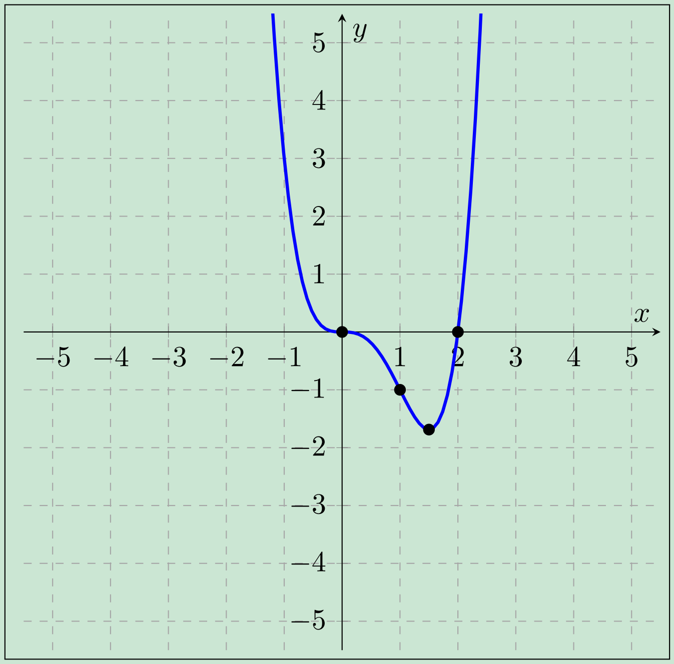
Now, let's plot these points on our graph, so we have points to trace for later.



Now, we can create our sign pattern table using our three sign patterns.

	$x<0$	$x=0$	$0<x<1$	$x=1$	$1<x<\frac{3}{2}$	$x=\frac{3}{2}$	$\frac{3}{2}<x<2$	$x=2$	$x>2$
y	$+$	0	$-$	-1	$-$	$-\frac{27}{16}$	$-$	0	$+$
y'	$-$	0	$-$	$-$	$-$	0	$+$	$+$	$+$
y''	$+$	0	$-$	0	$+$	$+$	$+$	$+$	$+$

Finally, we can refer to our 1st and 2nd derivative relationship table and connect the points to create our graph.



Ex 4.5.2: Find the sign patterns of y, y' and y'' and sketch $y = xe^{2x}$.

Sol 4.5.2: Again, just as in **Sol 4.5.1**, let us make our three sign patterns.
Zeroes:

$$xe^{2x} = 0 \implies x = 0 \qquad \to (0,\, 0)$$

$$\begin{array}{cccc} y & - & 0 & + \\ & \longleftarrow & \text{—} & \longrightarrow \\ x & & 0 & \end{array}$$

Extreme Values:

$$\frac{dy}{dx} = (2x+1)e^{2x} = 0 \implies x = \frac{1}{2} \qquad \to \left(-\frac{1}{2},\, -\frac{1}{2e}\right)$$

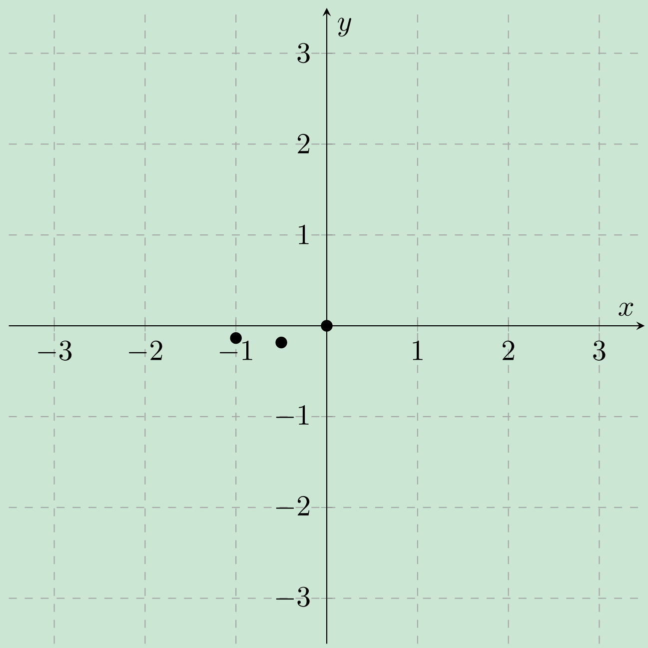
$$\begin{array}{cccc} y' & - & 0 & + \\ & \longleftarrow & \text{—} & \longrightarrow \\ x & & -\frac{1}{2} & \end{array}$$

Point of Inflection:

$$\frac{d^2y}{dx^2} = (4x+4)e^{2x} = 0 \implies x = -1 \qquad \to \left(-1,\, -\frac{1}{e^2}\right)$$

y''	$-$	0	$+$
x		-1	

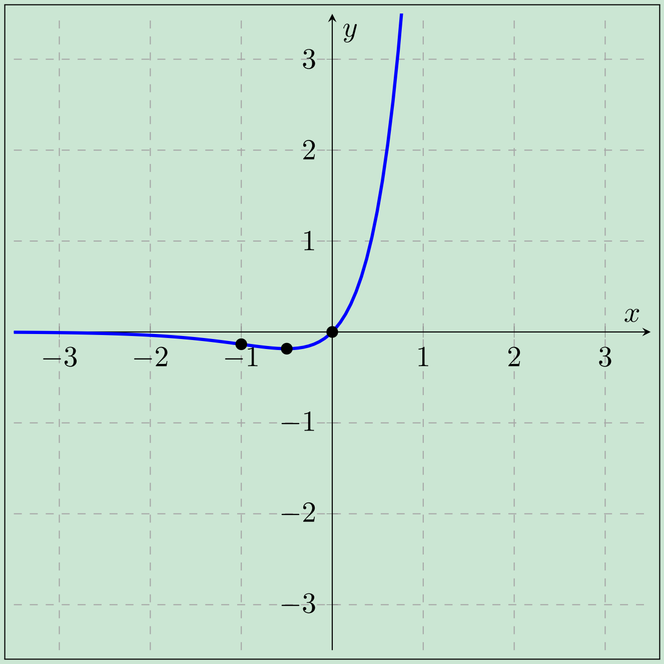
Now, let's plot the points that we've obtained.



We can now create our sign pattern table.

	$x<-1$	$x=-1$	$-1<x<-\frac{1}{2}$	$x=-\frac{1}{2}$	$-\frac{1}{2}<x<0$	$x=0$	$x>0$
y	$-$	$-\frac{1}{e^2}$	$-$	$-\frac{1}{2e}$	$-$	0	$+$
y'	$-$	$-$	$-$	0	$+$	$+$	$+$
y''	$-$	0	$+$	$+$	$+$	$+$	$+$

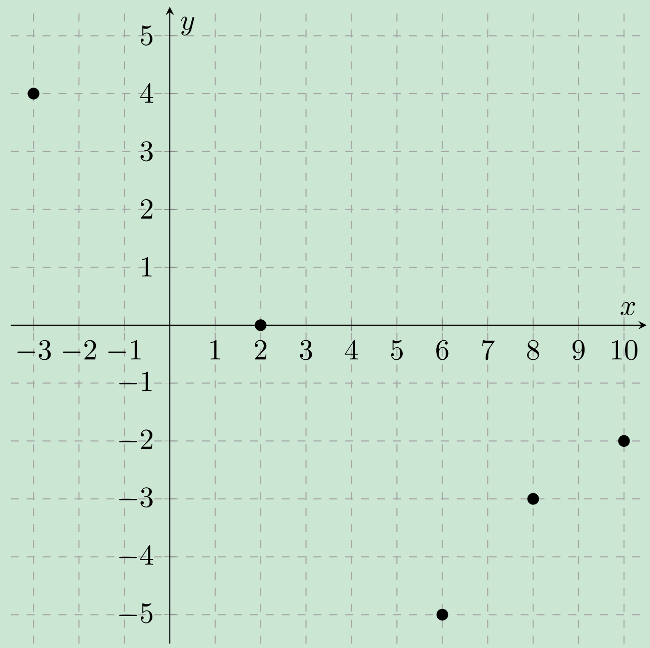
Finally, let's create a graph using our sign pattern table.



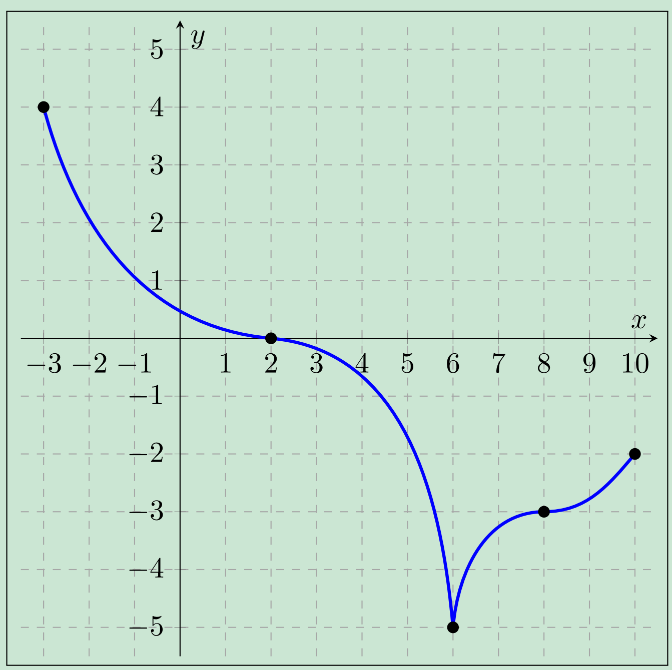
Ex 4.5.3: Sketch the graph of the function whose traits are given below. (Note that we are using interval notation in the problem, and the entries in the 'x' row are not coordinates)

x	-3	$(-3, 2)$	2	$(2, 6)$	6	$(6, 8)$	8	$(8, 10)$	10
$f(x)$	4	$+$	0	$-$	-5	$-$	-3	$-$	-2
$f'(x)$	DNE	$-$	0	$-$	DNE	$+$	$+$	$+$	DNE
$f''(x)$	DNE	$+$	0	$-$	DNE	$-$	0	$+$	DNE

Sol 4.5.3: Let's first plot the points that we are given, so we can connect the dots later.



Now, let's refer to the table given in the problem and draw the rest of the graph.

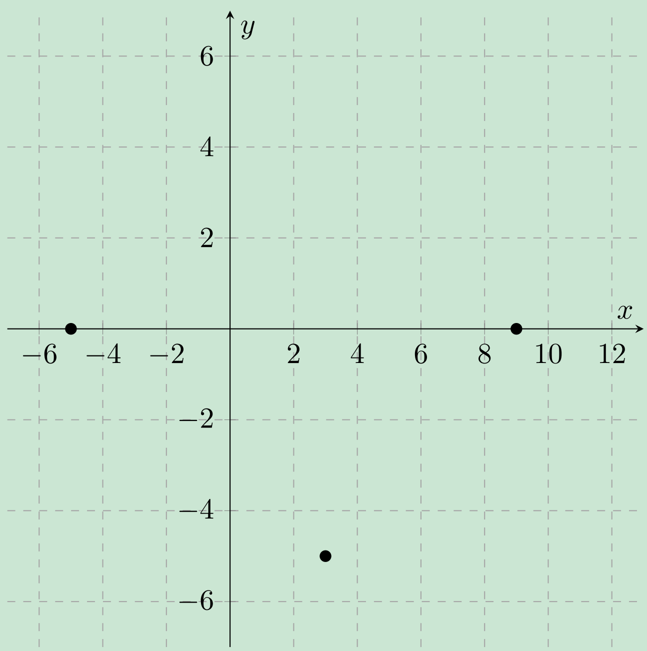


Note that in our previous example, we had some DNEs throughout the table. Remember that if a function has first or second derivative that does not exist at a certain point, that point is either an endpoint of the graph, or it is a "sharp" point on the graph.

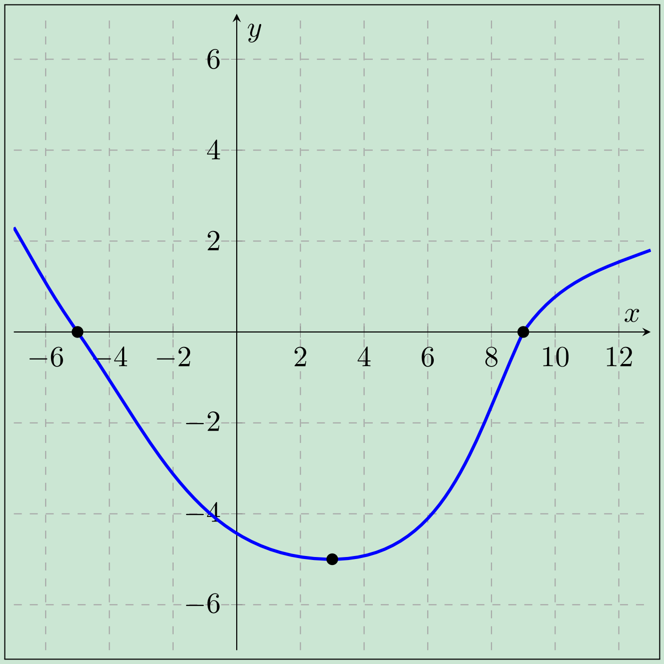
Ex 4.5.4: Sketch the graph of the function whose traits are given below.

	$x<-5$	$x=-5$	$-5<x<-3$	$x=3$	$3<x<9$	$x=9$	$x>9$
$f(x)$	$+$	0	$-$	-5	$-$	0	$+$
$f'(x)$	$-$	$-$	$-$	0	$+$	$+$	$+$
$f''(x)$	$+$	$+$	$+$	$+$	$+$	0	$+$

Sol 4.5.4: Plotting the points that we are given in the problem, we obtain this graph:



Now, let's connect the points with our desired graph.



y
6
4
2
x
−6
−4
−2
2
4
6
8
10
12
−2
−4
−6

4.5 Free Response Homework

Sketch these graphs using the sign patterns of the derivative.

1. $y = 3x^4 - 15x^2 + 7$

2. $y = x^3 + 5x^2 + 3x - 4$

3. $y = \dfrac{x+1}{x^2 - 2x - 3}$

4. $y = \dfrac{x^2 - 1}{x^2 + x - 6}$

5. $y = 5x^{\frac{2}{3}} - x^{\frac{5}{3}}$

6. $y = x^2\sqrt{4-x}$

7. $y = x^2 e^{-x}$

8. $y = \ln\left(x^2 + 4\right)$

9. The table of values and signs of $f(x)$ is given below. Sketch a graph of $f(x)$ on the domain represented on the table.

x	-3	$(-3, 2)$	2	$(2, 6)$	6	$(6, 8)$	8	$(8, 10)$	10
$f(x)$	4	$+$	0	$-$	-5	$-$	-3	$-$	-2
$f'(x)$	DNE	$-$	$-$	$-$	0	$+$	$+$	$+$	DNE
$f''(x)$	DNE	$+$	$+$	$+$	$+$	$+$	0	$-$	DNE

10. The table of values and signs of $f(x)$ is given below. Sketch a graph of $f(x)$ on the domain represented by the table.

x	-3	$(-3, 0)$	0	$(0, 3)$	3	$(3, 6)$	6	$(6, 8)$	8	$(8, 10)$	10
$f(x)$	0	$+$	3	$+$	0	$+$	3	$+$	0	$+$	3
$f'(x)$	DNE	$+$	0	$-$	DNE	$+$	DNE	$-$	0	$+$	DNE
$f''(x)$	DNE	$-$	$-$	$-$	DNE	0	DNE	$+$	$+$	$+$	DNE

11. The table of values and signs of $f(x)$ is given below. Sketch a graph of $f(x)$ on the domain represented on the table.

x	-3	$(-3, 0)$	0	$(0, 3)$	3	$(3, 6)$	6	$(6, 8)$	8	$(8, 10)$	10
$f(x)$	0	$+$	3	$+$	0	$-$	-3	$-$	0	$+$	3
$f'(x)$	DNE	$+$	DNE	$-$	$-$	$-$	DNE	$+$	0	$+$	DNE
$f''(x)$	DNE	$+$	DNE	$-$	DNE	$+$	DNE	$-$	0	$+$	DNE

12. The table of values and signs of $f(x)$ is given below. Sketch a graph of $f(x)$ on the domain represented on the table. [Hint: Be careful at $x = -2$]

x	-6	$(-6,\,-4)$	-4	$(-4,\,-2)$	-2	$(-2,\,0)$	0	$(0,\,2)$	2	$(2,\,4)$	4
$f(x)$	2	$+$	0	$+$	2	$+$	0	$-$	-2	$-$	0
$f'(x)$	DNE	$-$	0	$+$	DNE	$-$	$-$	$-$	DNE	$+$	0
$f''(x)$	DNE	$+$	$+$	$+$	DNE	$-$	0	$+$	DNE	$-$	$-$

13. A function f is continuous on the interval $x \in [-3,\,3]$ such that $f(-3) = 4$ and $f(3) = 1$. The functions f' and f'' have the properties given below. Sketch a graph of $f(x)$ on $x \in [-3,\,3]$.

x	$-3 < x < -1$	$x = -1$	$-1 < x < 1$	$x = 1$	$1 < x < 3$
$f'(x)$	$+$	DNE	$-$	0	$-$
$f''(x)$	$+$	DNE	$+$	0	$-$

14. A function f is continuous on the interval $x \in [-3,\,3]$ such that $f(-3) = 6$ and $f(3) = 1$. The functions f' and f'' have the properties given below. Sketch a graph of $f(x)$ on $x \in [-3,\,3]$.

x	$-3 < x < -1$	$x = -1$	$-1 < x < 1$	$x = 1$	$1 < x < 3$
$f'(x)$	$-$	0	$-$	DNE	$+$
$f''(x)$	$+$	0	$-$	DNE	$-$

15. A function f is continuous on the interval $x \in [-3,\,3]$ such that $f(-3) = 6$ and $f(3) = 1$. The functions f' and f'' have the properties given below. Sketch a graph of $f(x)$ on $x \in [-3,\,3]$.

x	$-3 < x < -1$	$x = -1$	$-1 < x < 1$	$x = 1$	$1 < x < 3$
$f'(x)$	$-$	DNE	$-$	0	$+$
$f''(x)$	$-$	DNE	$+$	3	$+$

16. A function f is continuous on the interval $x \in [-1,\,11]$ such that $f(-1) = 3$ and $f(11) = 4$. The functions f' and f'' have the properties given below. Sketch a graph of $f(x)$ on $x \in [-1,\,11]$.

x	$-1 < x < 4$	$x = 4$	$4 < x < 9$	$x = 9$	$9 < x < 11$
$f'(x)$	$-$	0	$+$	DNE	$+$
$f''(x)$	$+$	2	$+$	DNE	$+$

Sketch the possible graph of a function that satisfies the conditions indicated below.

17. Increasing from $(-\infty, 5) \cup (7, 10)$, decreasing from $(5, 7) \cup (10, \infty)$.

18. Increasing from $(-\infty, -3) \cup (5, \infty)$, decreasing from $(-3, 5)$. Concave up from $(-\infty, -2) \cup (2, \infty)$, concave down from $(-2, 2)$.

19. Decreasing from $(-\infty, -5) \cup (5, \infty)$, decreasing from $(-3, 5)$. Concave up from $(-\infty, -2) \cup (2, \infty)$, concave down from $(-2, 2)$.

20. Increasing and concave up from $(2, 4)$, decreasing and concave down from $(4, 7)$, increasing and concave up from $(7, 10)$, with a domain of $[2, 10]$.

21. Absolute minimum at $x = 1$, absolute maximum at $x = 3$, local minima at $x = 4$ and $x = 7$, local maximum at $x = 6$.

22. Absolute minimum at $x = 2$, absolute maximum at $x = 7$, local minima at $x = 4$ and $x = 6$, local maxima at $x = 3$ and $x = 5$.

23. Absolute minimum at $x = 7$, absolute maximum at $x = 1$, no local maxima or minima.

24. Absolute minima at $x = 1$ and $x = 7$, absolute maximum at $x = 4$, no local maxima or minima.

4.5 Multiple Choice Homework

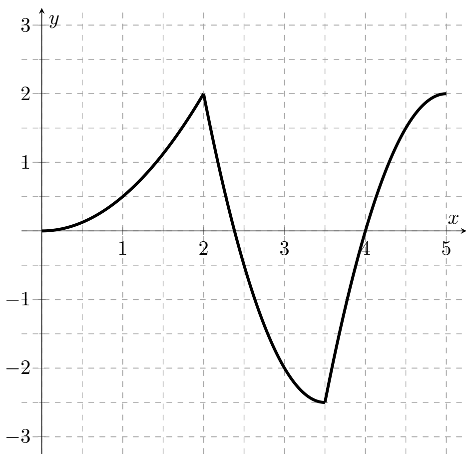
1. Which of the following graphs has these two sign patterns:

f': $+$ DNE $-$ DNE $+$

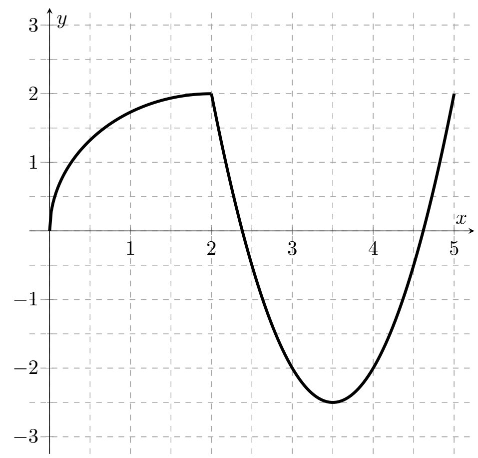
x: 2, 3.5

f'': $+$ DNE $+$ DNE $-$

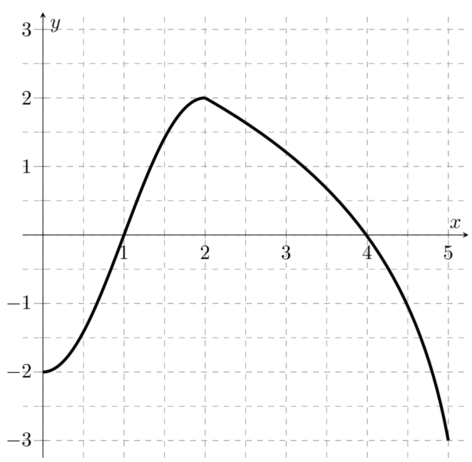
x: 2, 3.5



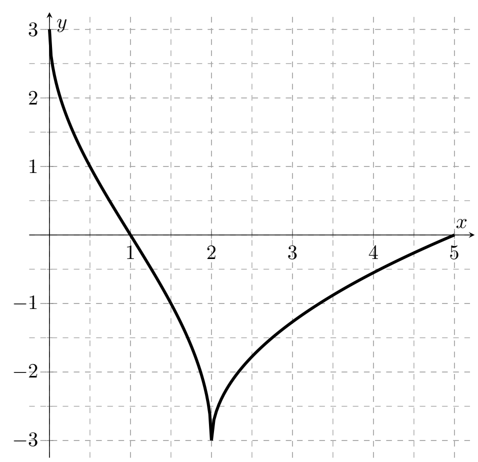
a)



b)



c)

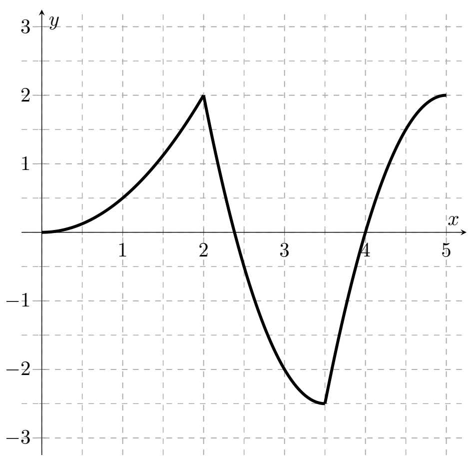


d)

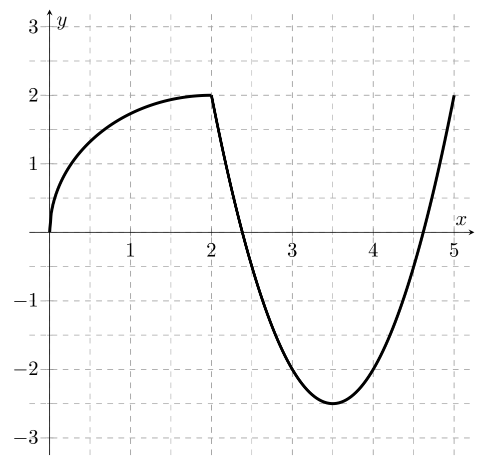
2. Which of the following graphs has these two sign patterns:

f'	$-$	DNE	$+$
x		2	

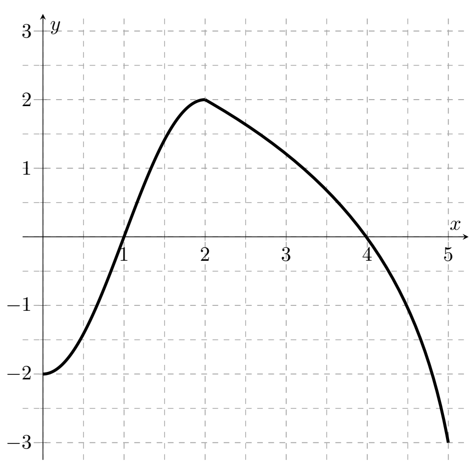
f''	$+$	0	$-$	DNE	$-$
x		1		2	



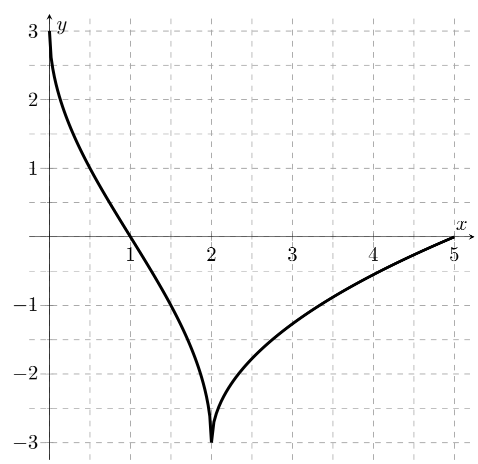
a)



b)



c)

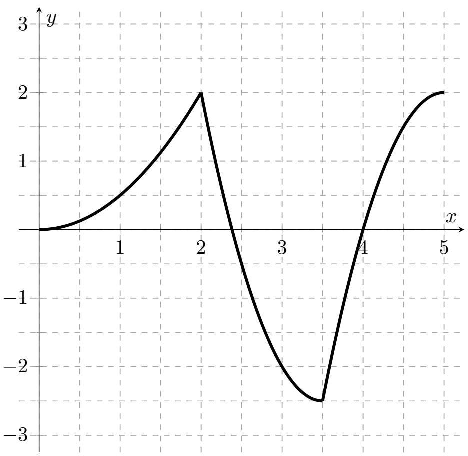


d)

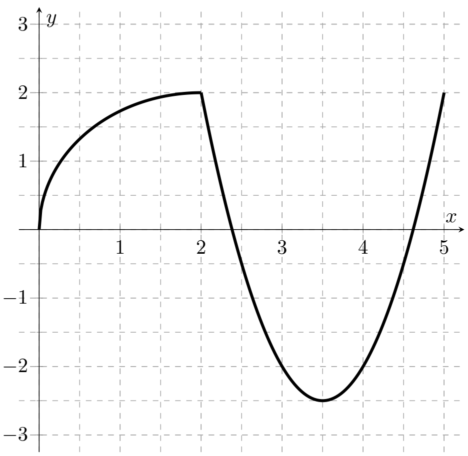
3. Which of the following graphs has these two sign patterns:

f'	$+$	DNE	$-$			
x		2				

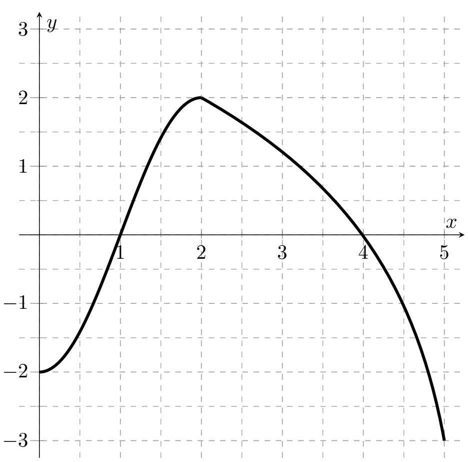
f''	$+$	0	$-$	DNE	$-$	
x		1		2		



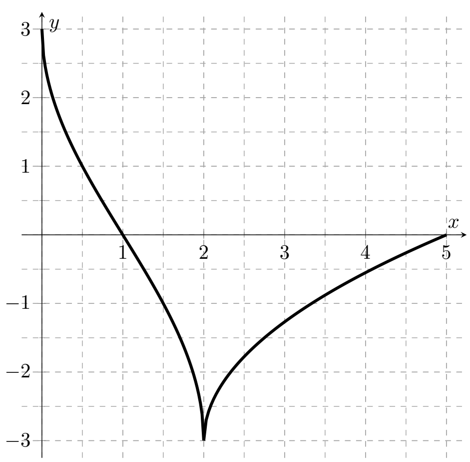
a)



b)



c)

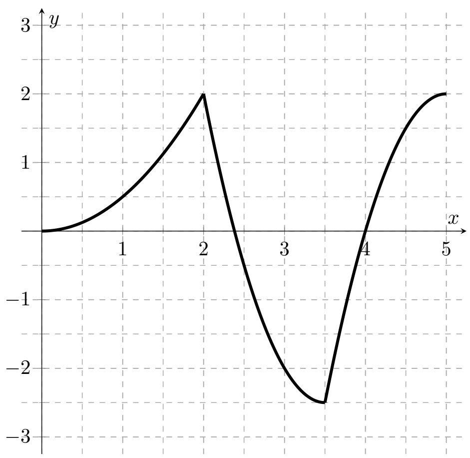


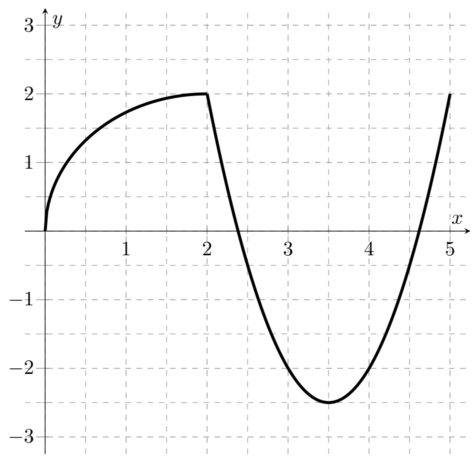
d)

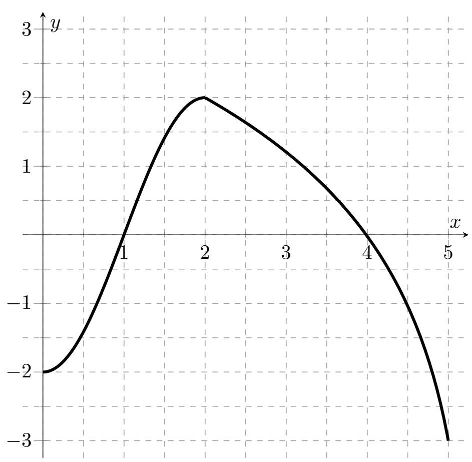
4. Which of the following graphs has these two sign patterns:

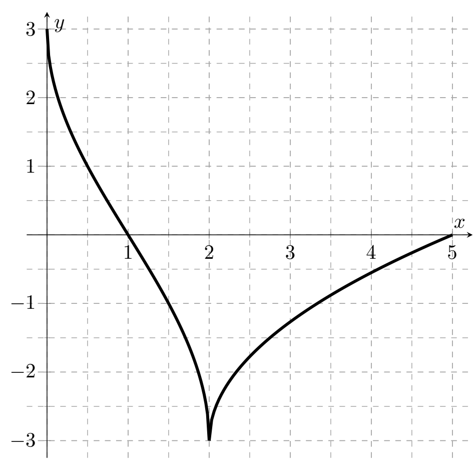
f'	$+$	DNE	$-$	DNE	$+$
x		2		3.5	

f''	$-$	DNE	$+$
x		2	

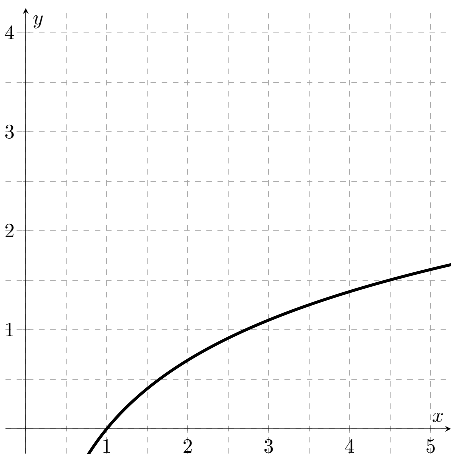
a) 

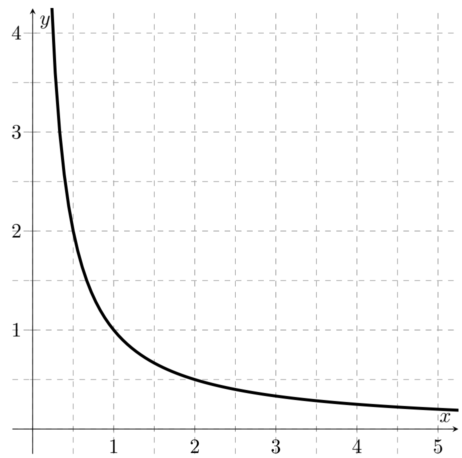
b) 

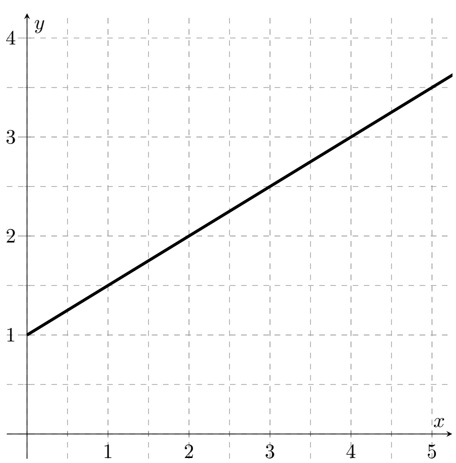
c) 

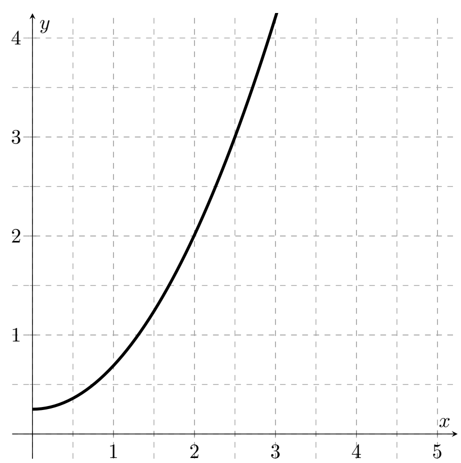
d) 

5. Which of the graphs of $y = g(x)$ below has $g'(x) < 0$ and $g''(x) > 0$?

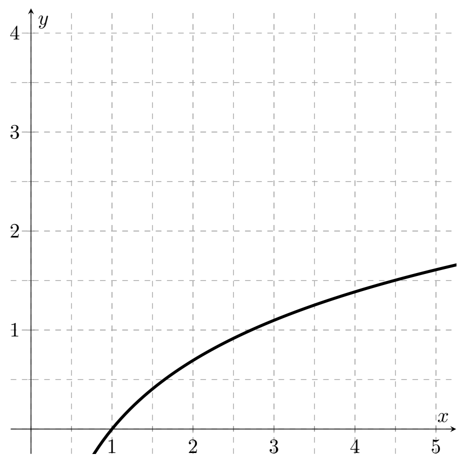
a) 

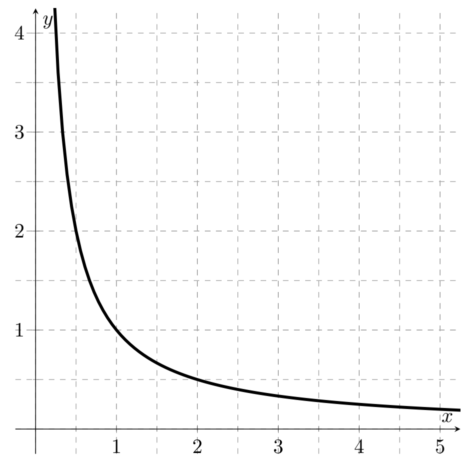
b) 

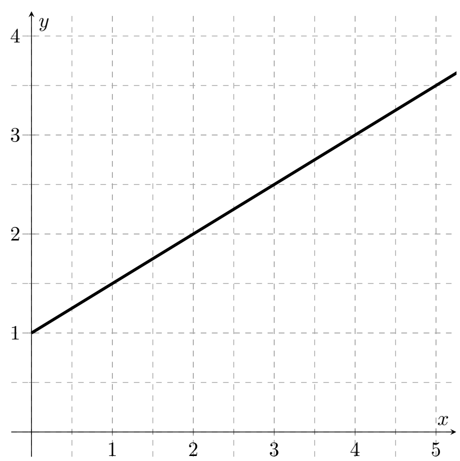
c) 

d) 

6. Which of the graphs of $y = g(x)$ below has $g'(x) > 0$ and $g''(x) > 0$?

a) 

b) 

c) 

d)